MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
2500 EXPLOSIVE MAGAZINE
EXPLOSIVES STORAGE BUILDING DESIGN CALCULATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of the Blast Resistant Explosive Building in Explosive Magazine Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Blast Analysis & Structural Design Verification of Explosive Building only.

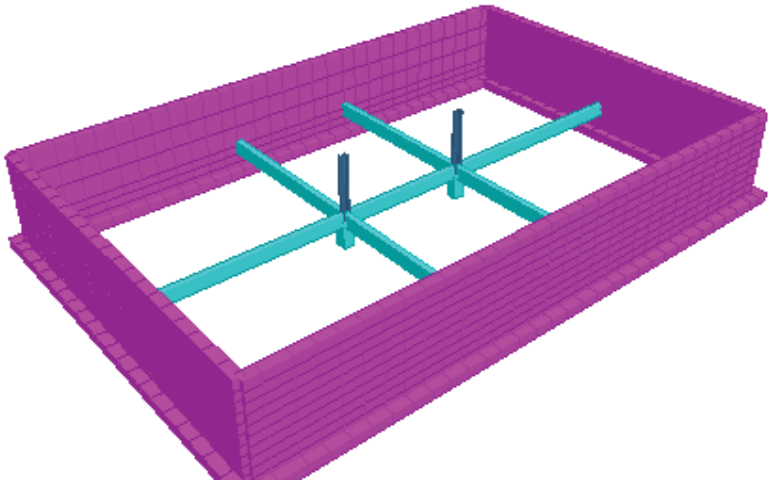
4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single-Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have RCC Walls and PEB Steel Framing Beams to support the Sheet Panel along with the Insulation material on Top (PEB Columns & Beams).The Building Analysis has been performed meeting design requirements which has been subjected to blast loading shown as per ASCE Report & Separate analysis was done by using STAAD Pro to generate the Reactions at the Base which are assumed to be partially fixed as per the Soil Modulus Reaction. The Maximum Foundation pressure has been checked by using STAAD Pro Design Software. However final Reactions shall be verified and checked for foundation design in IFC Stage.

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STAAD 3D MODEL

4.1. Codes

[1] ACI 318-08 Building Code Requirements.

[2] TM-5-1300-1900 Structures to resist the effects of accidental Explosions.

[3] ASCE Report - Manual on Design of Blast Resistant Buildings in Petrochemical Facilities.

[4] SBC-301-2007-Structural Loading & Forces.

[5] SBC-303-2007-Structural Soil & Foundations.

[6] SBC-304-2007-Structural Concrete Structures.

[7] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[8] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[9] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures Design of Blast Resistant Buildings in Petrochemical Facilities.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-C01.

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4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, Wall.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = **0.2 kN/m²**

5.1.2. Ground Floor Walls

- 300mm thk. RCC WALLS all around the Building by using Self weight command in STAAD Pro.

5.2. Roof Live Load

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat (Accessible) = **1.0 kN/m²**

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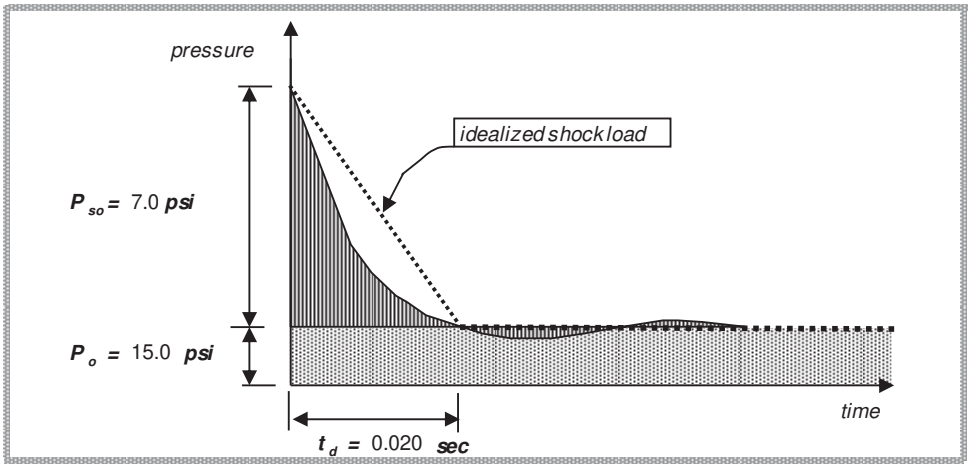
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5.3. Blast Loads

ASCE REPORT

Blast Wave Parameters

peak overpressure…….…….…….…….……	P_{so}=	7.0 psi
duration of positive phase…….……	t_d =	0.020 sec
ambient pressure…….……	P_o =	15.0 psi



(3.5) shock front velocity

$U = 1130 (1 + 0.058 P_{so})^{0.5} =$

$U = 1130 \times (1 + 0.058 \times 7)^{½} =$ 1340 ft/sec.

(3.4) Peak Dynamic Pressure - front wall

$q_{of} = 1.0 \times 2.5 [P_{so}^2 / (7 P_o + P_{so})] =$

$1.0 \times 2.5 \times [7^2 / (7 \times 15 + 7)] =$ 1.1 psi

(Note) # 3-8 Peak Dynamic Pressure - side wall, rear wall & roof

$q_{os} = -0.4 \times 2.5 [P_{so}^2 / (7 P_o + P_{so})] =$

$-0.4 \times 2.5 \times [7^2 / (7 \times 15 + 7)] =$ -0.44 psi

(3.6) length of blast wave $L_w = U t_{d=}$

$1339.9 \times 0.02 =$ 27 ft

impuls of the blast wave $0.5 \times P_{so} \times t_d =$

$0.5 \times 7 \times 0.02 =$ 0.07 $kN/m^2 sec$

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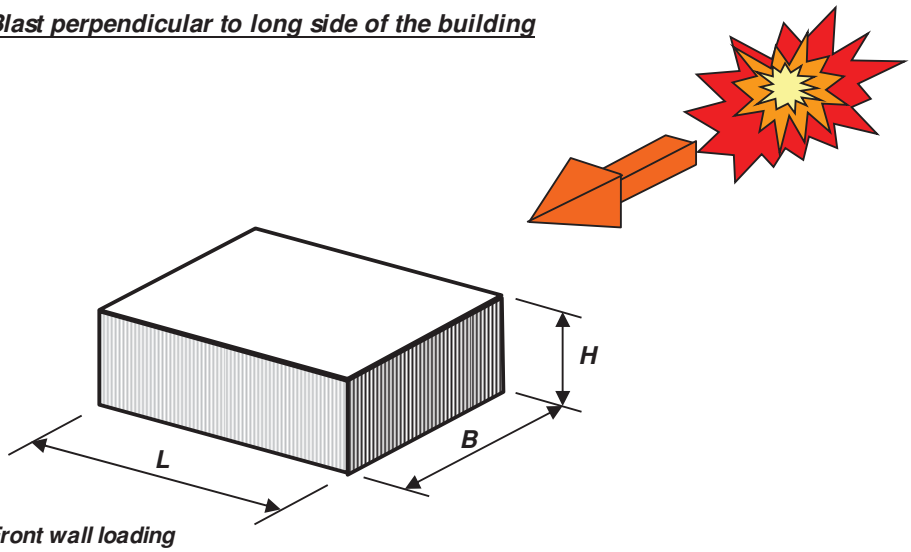
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Building geometry

(L = Blast Side of Building)

height.....	H =	3.0	m	=	9.84 ft
depth.....	B =	36.0	m	=	118.11 ft
length.....	L =	40.0	m	=	131.23 ft

Blast perpendicular to long side of the building



Front wall loading

(3.2) & (3.3) *peak reflected pressure* $P_r = (2 + 0.05 \times P_{so}) \times P_{so} =$

$(2 + 0.05 \times 7) \times 7 =$ 16.5 psi

S = clearing distance = min [L/2 ; H] Min (131.23 /2 & 9.84) = 10 ft

reflected pressure clearig time $t_c = 3 \times S / U =$ $3 \times 9.84 / 1339.9 =$ 0.022 sec

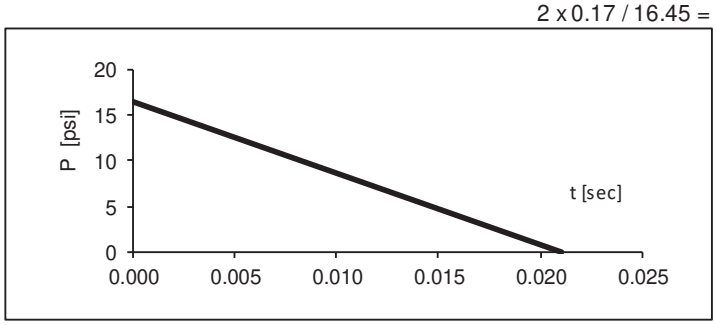
stagnation pressure $P_s = P_{so} + q_{of} =$ $7 + 1.09 =$ 8.1 psi

impulse load $I_w = 0.5 \times (P_r - P_s) \times t_c + 0.5 \times P_s \times t_d =$

$(0.5 \times (16.45 - 8.09) \times (0.022)) + (0.5 \times 8.09 \times (0.02)) =$ 0.173 psi-sec

Effective (corresponding) duration $t_e = 2 \times I_w / P_r =$

$2 \times 0.17 / 16.45 =$ 0.021 sec



Force (psi)	Time (sec)
16.5	0.000
0.0	0.021

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Side wall loading

The side wall is the same as the front wall, spanning vertically from foundtion to roof. Because the highest load on the front wall, a side wall analysis would only be necessory to check the interaction of in-plane and out-of-plane shear wall forces. This calculation will be for a wall segment, L1, 1 ft. wide

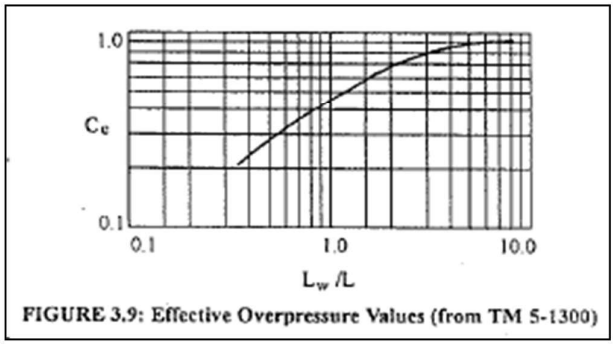


FIGURE 3.9: Effective Overpressure Values (from TM 5-1300)

Reduction factor C_e for L_w / B = (B = 1 ft, unit length of side wall)

26.8 / 1 = 27 ∴ C_e = 1.00

Equivalent peak overpressure

$P_a = C_e P_{so} + q_{os}$ =

(1 x 7) + (-0.44) = 6.6 psi

rise time

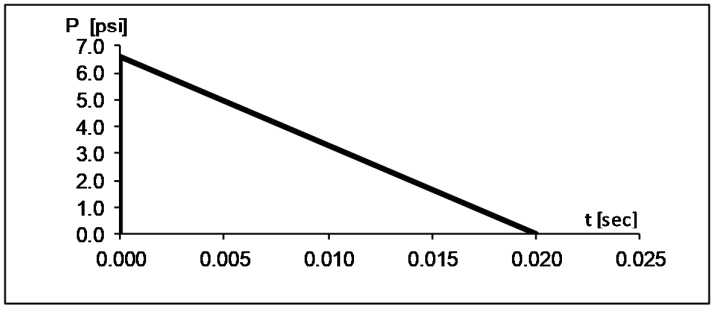
$t_{rise}^{side} = B / U$ =

1 / 1339.9 = 0.0007 sec ≈ 0

total duration

$t_{rise}^{side} + t_d$ =

0 + 0.02 = 0.020 sec



Force (psi)	Time (sec)
0.0	0.000
6.6	0.000
0.0	0.020

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Average Overpressure on Side wall

Reduction factor C_e for L_w / B = (B = Length of side wall)

26.8 / 118.11 = 0.227 $\therefore$ C_e = 0.21

Equivalent peak overpressure $P_a = C_e P_{so} + q_{os}$ =

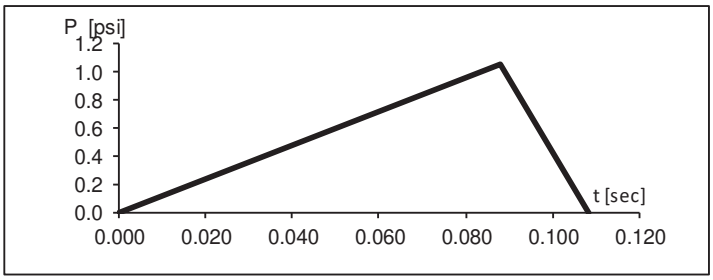
(0.21 x 7) + (-0.44) = 1.1 psi

rise time $t_{rise}^{side} = B / U$ =

118.11 / 1339.9 = 0.088 sec

total duration $t_{rise}^{side} + t_d$ =

0.08815 + 0.02 = 0.108 sec



Force (psi)	Time (sec)
0.0	0.000
1.1	0.088
0.0	0.108

From above figure - If an average overpressure over the entire side wall is needed, the value of L1 would then be the lenth of the building. The value of Ce would then be less than one and thus the value of Pa. The rise time would become significant

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Rear Wall Load

Rear wall blast load lags that for the front wall by the time for blast wave to travel depth of the building. The rise time is also influenced by spillover from roof and side walls and depends on the building length or height. Minimum value of (L/2 and H) governs rise time. The effective rear wall overpressure is almost identical to side-on pressure.

time deley to reach rear wall

$t_{deley} = B / U =$

$118.11 / 1339.9 =$ 0.088 sec

rise time

$t_{rise}^{rear} = S / U =$

$9.84 / 1339.9 =$ 0.007 sec

total time to peak

$t_{total} = t_{deley} + t_{rise}^{rear} =$

$0.0881 + 0.0073 =$ 0.095 sec

total duration time

$t_d^{rear} = t_{total} + t_d =$

$0.0955 + 0.02 =$ 0.115 sec

total positive phase duration…….....…. $t_o = t_d^{rear} - t_{deley} =$

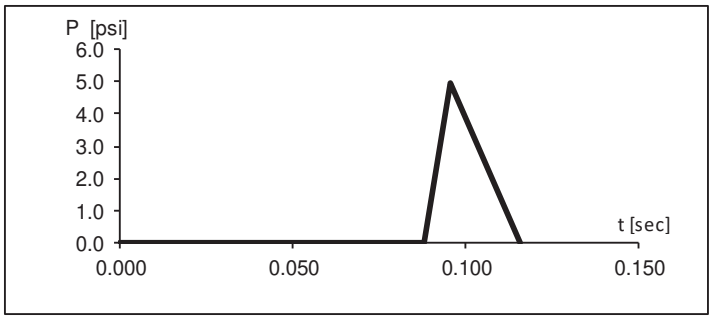
$0.1155 - 0.0881 =$ 0.027 sec

reduction factor C_e for $L_w/min[L/2,H]=$ 2.7 $\therefore$ $C_e =$ 0.77

peak overpressure

$P_a = C_e P_{so} + q_{os} =$

$(0.77 \times 7) + (-0.44) =$ 4.9 psi



Force (psi)	Time (sec)
0.0	0.000
0.0	0.088
4.9	0.095
0.0	0.115

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6. DESIGN COMBINATIONS

Loads combinations shall be in accordance to IBC 2003, Section 1605 and ASCE 7-02, Section 2.3 & 2.4 and ASCE, Blast Resistant Building in Petrochemical Facilities Section 7.2.4.

The Following Load Combinations shall be applied for foundation stability check and concrete structure design:

Ultimate Load Combinations

1.2D + 1.6L + B

Allowable Load Combinations

D + L + B

Where:

D = Dead Load

L = Live Load

B = Blast Load

Note: Wind and Seismic Loads have been neglected as Blast Loading is critical and will be acting independent of these accidental loads

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7. DESIGN ON FRONT WALL (BLAST ANALYSIS)

Dynamic Parameters of Reinforced Concrete

ASCE Report Table 5.A.1

Strength Increase Factor (SIF)

material	factor
reinforcing bars	1.10
concrete	1.00

table 1

ASCE Report Table 5.A.2

Dynamic Icrease Factor (DIF)

stress type	reinforcing bars		concrete
	f_{dy}/f_y	f_{du}/f_u	f_{dc}/f_c
flexure	1.17	1.05	1.19
compression	1.10	1.00	1.12
diagonal tension	1.00	1.00	1.00
direct shear	1.10	1.00	1.10
bond	1.17	1.05	1.00

table 2

note:

the above tables are based on the Technical Manual TM 5-1300, Structures to Resist the Effect of Accidental Explosions, published by the Department of Army Navy, and Air Force, Washington DC,1990

Material properties

Grade of Concrete (f_{ck}) =	35 N/mm^2	5	ksi
Grade of Steel (f_y) =	415 N/mm^2	60	ksi
Con. tensile strength = 0.7 $\sqrt{f_{ck}}$	4.1 N/mm^2	1.6	ksi
Ec = 5000 $\sqrt{f_{ck}}$	29580 N/mm^2	4290	ksi
Es =	200000 N/mm^2	29008	ksi
Density of concrete (γ) =	25 kN/m^3	159	lb / ft^3
Modular ratio (n = Es/Ec) =	6.76	6.76	

Wall Section - Front Wall

Geometrical properties, trial size

Height of wall	H =	3.00	m	9.84	ft
width of the section	b =	1	ft	12	in
overall depth	D =	300	mm	12	in
Clear Cover to reinforcement		30	mm	1.20	in
effective depth	d =	260	mm	10.41	in

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Reinforcement in Front wall

Vertical bar (outside the horizontal bar)

Dia	Spc.	Area	Dia	Area	Spc.	Area
mm	mm c/c	mm^2 / ft	in	in^2	in c/c	in^2 / ft
20	125	766	0.787	0.49	5	1.18

	SIF	DIF
Concrete	1	1.19
Steel	1.1	1.17

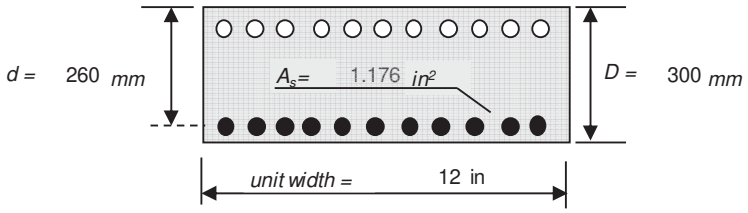
f_{dy} = (SIF) x (DIF) x f_y = (1.1) x (1.17) x 60 = 77.20 ksi

f_{dc} = (SIF) x (DIF) x f_{ck} = (1) x (1.19) x 5 = 5.95 ksi

A_s - Area of Reio. provided = 0.49(bar area) / 5 (spc.) x 12" = 1.176 in^2

Min. reo. Ratio = min [200 / fdy ; 3√fdc / fdy] = min[0.0026 ; 0.003] = 0.0026

ρ = Reo. Ratio provided = 1.176 / (12 x 10.4) = 0.0094 > As min, Ok



Compute Bending Resistance:-

For dynamic Bending:

Depth of Neutral Axis = a = A_s x f_{dy} / (0.85 x f'_{dc} x b) =

1.18 x 77.2 / (0.85 x 5.95 x 12) = 1.50 in

Plastic moment = M_p = M_n = A_s x f_{dy} x [d - a / 2] =

1.18 x 77.2 x [10.41 - 1.5 / 2] = 877 in-kip

R_b = 8 M_p / L = 8 x 877 / (9.84 x 12) = 59.40 kips

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Compute Shear Resistance:-

For dynamic Shear:

	SIF	DIF
Concrete	1	1

f^{v}_{dc} = (SIF) x (DIF) x f_{cu} = (1) x (1) x 5 = 5.00 ksi

Nominal Shear strength = V_n = 2 x √ f'c x b x d =
2 x SQRT(5 x 1000) x 12 x 10.41 / 1000 = 17.67 kips

Critical section for shear is at distance 'd' from face of support
$R_s = V_n L / (0.5 L - d)$ =
17.67 x 9.84 x 12 / ((0.5 x 9.84 x 12) - 10.41) = 42.90 kips

Compute SDOF Equivalent System:-

Because Rb > Rs, Shear Control
Governing Resistance, R_u = min(R_b & R_s) = ***42.90*** kips

Gross Moment of Inertia = $I_g = b \times D^3 / 12$ =
12 x 12 ^3 / 12 = 1728 in^4

Cracked moment of inertia

n x A_s = 6.76 x 1.176 = 7.95 in^2

$C = (-n A_s + \sqrt{ (n A_s (n A_s + 2 b d))) } / b$
((-7.95) + (SQRT (7.95 x (7.95 + 2 x 12 x 10.41)))) / 12 = 3.11 in

Cracked Moment of inertia = $I_{cr} = (b \times C^3 / 3) + (n \times A_s \times (d - C)^2)$ =
(12 x 3.11 ^3 / 3) + (7.95 x (10.41 - 3.11) ^2) = 544 in^4

Average moment of Inertia = I_a = (Ig + Icr) / 2 = 1136 in^4

Effective Stiffness = $K = 384 E I / 5 L^3$ =
384 x 4290 x 1136 / (5 x (9.84 x 12) ^3) = 227.16 kip/in

Yield Deflecition = $y_e = R_u / K$ = 42.9 / 227.16 = 0.19 in

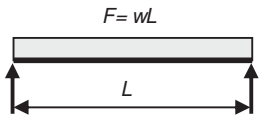
Acceleration due to gravity = g = 386 in/sec^2

Beam Mass = $M = \gamma b D H / g$ =
(159 / 1000) x (12 / 12) x (12 / 12) x (9.84) / 386 = 0.00405 $k\text{-}sec^2 / in$

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Ref Table 6.1

Strain Range	K_L	K_M^{lumped}	$K_M^{uniform}$	R_b	K	V_{left}	V_{right}
Elastic	0.64	n/a	0.50	$8M_{pc}/L$	$384EI/5L^3$	$0.39R+0.11F$	$V_{right}=V_{left}$
Plastic	0.50	n/a	0.33	$8M_{pc}/L$	0	$0.38R_u+0.12F$	$V_{right}=V_{left}$

Stage	Mass Factor K_M	Load Factor K_L	K_{LM} = K_M / K_L	Average K_{LM}
Elastic	0.5	0.64	0.78	0.72
Plastic	0.33	0.5	0.66	

Equivalent Mass = $M_e = K_{LM} \times M$ = 0.72 x 0.00405 = 0.00292 k-sec^2 / in

Period of vibration = $t_n = 2 \times \pi \times \sqrt{(M_e / K)}$ =
2 x pi x SQRT (0.00292 / 227.16) = 0.023 sec

Damping ratio = ξ = 0 %

Damping constant = $c = 2 \times \xi \times \sqrt{(K \times M_e)}$ =
2 x 0 x SQRT (227.16 x 0.00292) / 100 = 0.000 k-sec/in

Blast load duration (For Front wall) = t_d = 0.0210 sec

Required Time step = min (t_d / 10 & t_n / 10) = 0.0021 sec

Time step considered for Numeriacl Integration = 0.0020

table above Support Reaction = V (t) = 0.385 R + 0.115 F

Blast Pressure on Front wall = P_r = 16.50 psi

Loading on Front wall = P_r H b = 16.5 x 9.84 x 12 x 12 / 1000 = 23.4 kips

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Input Data summary for Numerial Integration

Force (kips)	Time (sec)
23.4	0
0	0.0210

K =	227.16 kip/in
m = M_e =	0.00292 kips-sec^2 / in
c =	0.000 kips-sec/in
Rt =	42.90 kips
Rc =	-42.90 kips

Numerical Integration solution (Front Wall)

time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.000	23.40	0.000	0.00	8013.7	0.00	2.69
0.002	21.17	0.015	14.12	6102.8	3.35	3.72
0.004	18.94	0.053	22.61	2386.4	11.97	6.79
0.006	16.71	0.100	22.96	-2035.9	22.66	10.65
0.008	14.49	0.139	15.06	-5856.0	31.59	13.83
0.010	12.26	0.156	1.26	-7943.6	35.45	15.06
0.012	10.03	0.143	-14.36	-7681.2	32.46	13.65
0.014	7.80	0.100	-27.19	-5146.5	22.83	9.69
0.016	5.57	0.039	-33.42	-1089.2	8.75	4.01
0.018	3.34	-0.028	-31.22	3290.3	-6.26	-2.03
0.020	1.11	-0.081	-21.24	6696.4	-18.44	-6.97
0.022	0.00	-0.109	-6.06	8484.3	-24.77	-9.54
0.024	0.00	-0.104	10.55	8125.0	-23.72	-9.13
0.026	0.00	-0.069	24.04	5362.1	-15.66	-6.03
0.028	0.00	-0.013	30.42	1012.8	-2.96	-1.14
0.030	0.00	0.047	27.79	-3636.0	10.62	4.09
0.032	0.00	0.093	16.95	-7209.2	21.05	8.10
0.034	0.00	0.111	1.09	-8649.7	25.26	9.72
0.036	0.00	0.097	-15.09	-7531.2	21.99	8.47
0.038	0.00	0.054	-26.81	-4184.8	12.22	4.70
0.040	0.00	-0.005	-30.59	399.6	-1.17	-0.45
0.042	0.00	-0.063	-25.33	4865.8	-14.21	-5.47
0.044	0.00	-0.101	-12.57	7892.6	-23.05	-8.87
0.046	0.00	-0.110	3.91	8584.4	-25.07	-9.65
0.048	0.00	-0.087	19.23	6736.7	-19.67	-7.57
0.050	0.00	-0.037	28.86	2896.0	-8.46	-3.26
0.052	0.00	0.023	29.96	-1801.4	5.26	2.03
0.054	0.00	0.077	22.19	-5965.9	17.42	6.71
0.056	0.00	0.108	7.86	-8365.5	24.43	9.40
0.058	0.00	0.107	-8.80	-8290.2	24.21	9.32

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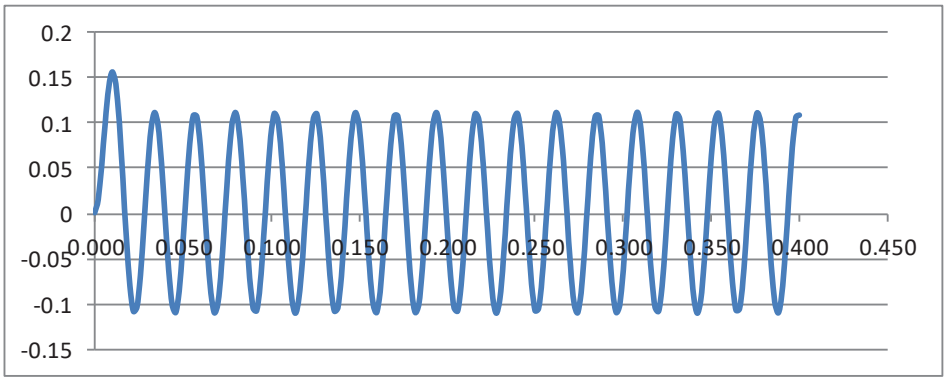
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time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.060	0.00	0.074	-22.85	-5762.5	16.83	6.48
0.062	0.00	0.020	-30.14	-1529.9	4.47	1.72
0.064	0.00	-0.041	-28.52	3155.2	-9.21	-3.55
0.066	0.00	-0.089	-18.46	6906.9	-20.17	-7.76
0.068	0.00	-0.111	-2.93	8615.3	-25.16	-9.69
0.070	0.00	-0.100	13.46	7775.0	-22.70	-8.74
0.072	0.00	-0.060	25.87	4634.6	-13.53	-5.21
0.074	0.00	-0.002	30.62	123.1	-0.36	-0.14
0.076	0.00	0.057	26.32	-4424.8	12.92	4.97
0.078	0.00	0.099	14.23	-7663.7	22.38	8.62
0.080	0.00	0.111	-2.07	-8635.4	25.22	9.71
0.082	0.00	0.091	-17.75	-7052.4	20.59	7.93
0.084	0.00	0.043	-28.19	-3383.1	9.88	3.80
0.086	0.00	-0.017	-30.29	1287.1	-3.76	-1.45
0.088	0.00	-0.072	-23.42	5576.5	-16.28	-6.27
0.090	0.00	-0.106	-9.63	8216.2	-23.99	-9.24
0.092	0.00	-0.108	7.01	8425.2	-24.60	-9.47
0.094	0.00	-0.079	21.58	6141.8	-17.93	-6.90
0.096	0.00	-0.026	29.76	2041.4	-5.96	-2.29
0.098	0.00	0.034	29.14	-2662.9	7.78	2.99
0.100	0.00	0.085	19.90	-6579.4	19.21	7.40
0.102	0.00	0.110	4.77	-8549.5	24.96	9.61
0.104	0.00	0.103	-11.77	-7990.4	23.33	8.98
0.106	0.00	0.065	-24.83	-5067.4	14.80	5.70
0.108	0.00	0.008	-30.54	-645.3	1.88	0.73
0.110	0.00	-0.051	-27.22	3967.7	-11.59	-4.46
0.112	0.00	-0.095	-15.84	7406.9	-21.63	-8.33
0.114	0.00	-0.111	0.22	8654.9	-25.27	-9.73
0.116	0.00	-0.094	16.22	7342.5	-21.44	-8.25
0.118	0.00	-0.050	27.42	3857.9	-11.27	-4.34
0.120	0.00	0.010	30.51	-768.0	2.24	0.86
0.122	0.00	0.066	24.57	-5166.7	15.09	5.81
0.124	0.00	0.103	11.37	-8036.9	23.47	9.04
0.126	0.00	0.110	-5.20	-8529.5	24.91	9.59
0.128	0.00	0.084	-20.23	-6498.8	18.98	7.31
0.130	0.00	0.033	-29.27	-2545.5	7.43	2.86
0.132	0.00	-0.028	-29.66	2160.8	-6.31	-2.43
0.134	0.00	-0.080	-21.27	6227.9	-18.19	-7.00
0.136	0.00	-0.109	-6.59	8452.6	-24.68	-9.50
0.138	0.00	-0.105	10.04	8176.6	-23.88	-9.19

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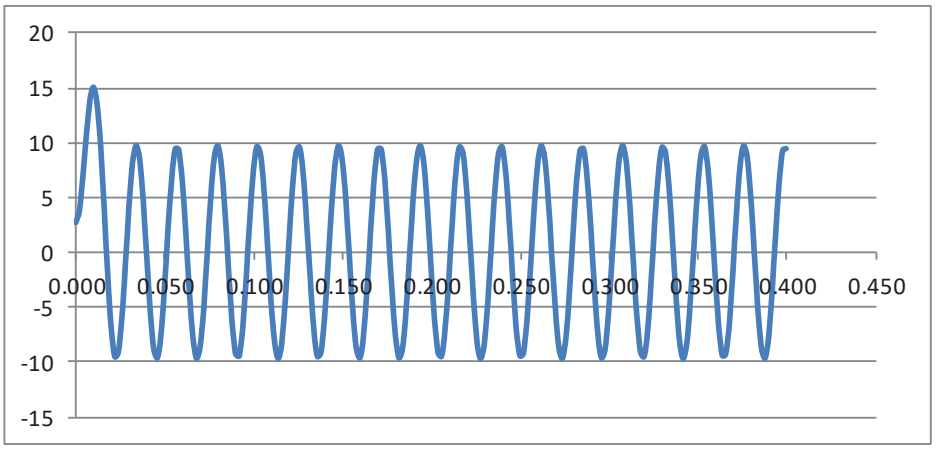
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Displacement versus time

max positive deflection y_{max} = 0.156 in @ t = 0.010 sec
max rebound deflection y_{reb} = -0.109 in @ t = 0.022 sec

Plastic Deformation = $y_p = y_m - y_e$ = 0.156 - 0.19 = 0.000 in
Rebound elastic deformation = $y_{reb} - y_p$ = 0.109 - 0 = 0.109 in



Reaction versus time

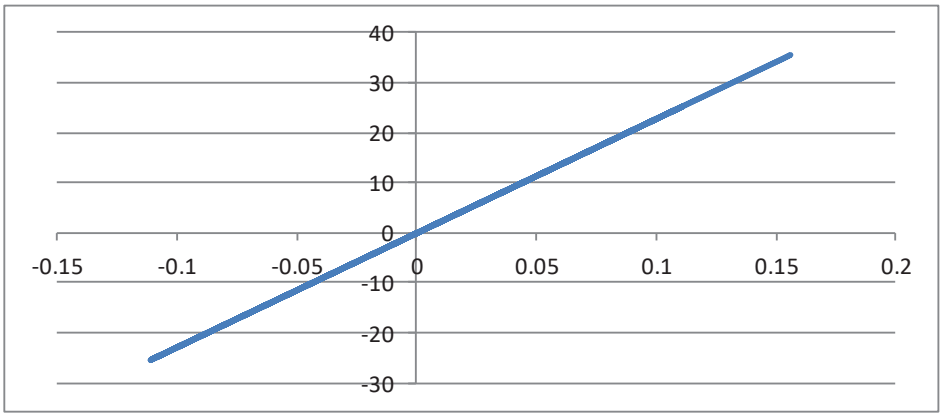
max dynamic reaction R_{dyn} = 15.059 kip/ft@ t= 0.010 sec
max rebound reaction R_{reb} = -9.538 kip/ft@ t= 0.022 sec

Support rotation check
$\theta = \arctan(2\ y_m / H)$ = Atan(2 x 0.156 / (9.84 x 12) = 0.15 deg
OK, rotation within 2 degree limit

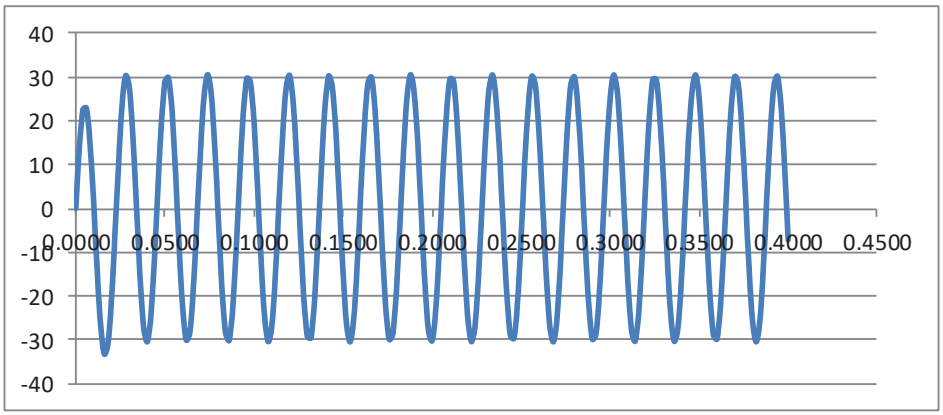
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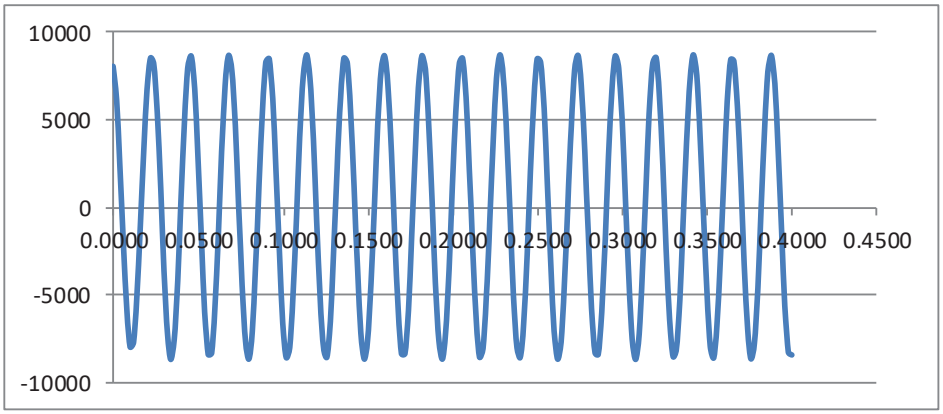
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Dynamic Resistance versus Displacement



Velocity versus Time



Acceleration versus Time

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8. DESIGN ON SIDE WALL (OUT OF PLANE)

Material properties

Grade of Concrete (f_{ck}) =	35 N/mm^2	5	ksi
Grade of Steel (f_y) =	415 N/mm^2	60	ksi
Con. tensile strength = 0.7 $\sqrt{f_{ck}}$	4 N/mm^2	1.6	ksi
Ec = 5000 $\sqrt{f_{ck}}$	29580 N/mm^2	4290	ksi
Es =	200000 N/mm^2	29008	ksi
Density of concrete (γ) =	25 kN/m^3	159	lb / ft^3
Modular ratio (n = Es/Ec) =	6.76	6.76	

Wall Section - Side Wall

Geometrical properties, trial size

Height of wall	H =	3.00	m	9.84 ft
width of the section	b =	1.00	ft	12 in
overall depth	D =	300	mm	12 in
Clear Cover to reinforcement		30	mm	1.20 in
effective depth	d =	260	mm	10.41 in

Reinforcement in Rear wall

Vertical bar (outside the horizontal bar)

Dia	Spc.	Area	Dia	Area	Spc.	Area
mm	mm c/c	mm^2 / ft	in	in^2	in c/c	in^2 / ft
20	125	766	0.787	0.49	5	1.18

	SIF	DIF
Concrete	1	1.19
Steel	1.1	1.17

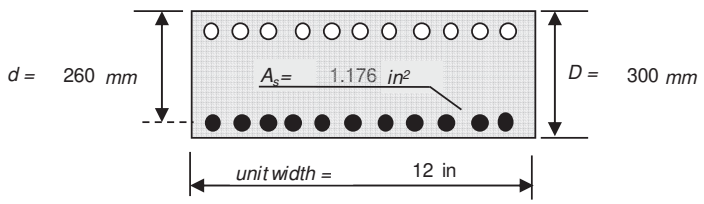
f_{dy} = (SIF) x (DIF) x f_y = (1.1) x (1.17) x 60 = 77.20 ksi

f_{dc} = (SIF) x (DIF) x f_{ck} = (1) x (1.19) x 5 = 5.95 ksi

A_s - Area of Reio. provided = 0.49(bar area) / 5 (spc.) x 12" = 1.176 in^2

Min. reo. Ratio = min [200 / fdy ; 3$\sqrt{}$fdc / fdy] = min[0.0026 ; 0.003] = 0.0026

ρ = Reo. Ratio provided = 1.176 / (12 x 10.4) = 0.0094 > As min, Ok



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Compute Bending Resistance:-

For dynamic Bending:

Depth of Neutral Axis = a = $A_s \times f_{dy} / (0.85 \times f'_{dc} \times b) =$
$1.18 \times 77.2 / (0.85 \times 5.95 \times 12) =$ 1.50 in

Plastic moment = $M_p = M_n =$ $A_s \times f_{dy} \times [d - a / 2] =$
$1.18 \times 77.2 \times [10.41 - 1.5 / 2] =$ 877 in-kip

$R_b = 8 M_p / L =$ $8 \times 877 / (9.84 \times 12) =$ 59.40 kips

Compute Shear Resistance:-

For dynamic Shear:

	SIF	DIF
Concrete	1	1

$f^v_{dc} = (SIF) \times (DIF) \times f_{cu} =$ $(1) \times (1) \times 5 =$ 5.00 ksi

Nominal Shear strength = $V_n =$ $2 \times \sqrt{f'c} \times b \times d =$
$2 \times SQRT(5 \times 1000) \times 12 \times 10.41 / 1000 =$ 17.67 kips

Critical section for shear is at distance 'd' from face of support
$R_s = V_n L / (0.5 L - d) =$
$17.67 \times 9.84 \times 12 / ((0.5 \times 9.84 \times 12) - 10.41) =$ 42.90 kips

Compute SDOF Equivalent System:-

Because Rb > Rs, Shear Control

Governing Resistance, $R_u = \min(R_b \& R_s) =$ ***42.90*** kips

Gross Moment of Inertia = $I_g = b \times D^3 / 12 =$
$12 \times 12 \text{ ^3} / 12 =$ 1728 in^4

Cracked moment of inertia

$n \times A_s =$ $6.76 \times 1.176 =$ 7.95 in^2

$C = (-n A_s + \sqrt{} (n A_s (n A_s + 2 b d))) / b$
$((-7.95) + (SQRT (7.95 \times (7.95 + 2 \times 12 \times 10.41)))) / 12 =$ 3.11 in

Cracked Moment of inertia = $I_{cr} = (b \times C^3 / 3) + (n \times A_s \times (d - C)^2) =$
$(12 \times 3.11 \text{ ^3} / 3) + (7.95 \times (10.41 - 3.11) \text{ ^2}) =$ 544 in^4

Average moment of Inertia = $I_a = (Ig + Icr) / 2 =$ 1136 in^4

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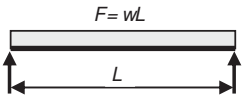
Effective Stiffness = K = 384 E I / 5 L^3 =
384 x 4290 x 1136 / (5 x (9.84 x 12) ^3) = 227.16 kip/in

Yield Deflecition = $y_e = R_u / K$ = 42.9 / 227.16 = 0.19 in

Acceleration due to gravity = g = 386 in/sec^2

Beam Mass = M = γ b D H / g =
(159 / 1000) x (12 / 12) x (12 / 12) x (9.84) / 386 = 0.00405 $k\text{-}sec^2$ / in

Ref Table 6.1



Strain Range	K_L	K_M^{lumped}	$K_M^{uniform}$	R_b	K	V_{left}	V_{right}
Elastic	0.64	n/a	0.50	$8M_{pc}/L$	$384EI/5L^3$	0.39R+0.11F	$V_{right}=V_{left}$
Plastic	0.50	n/a	0.33	$8M_{pc}/L$	0	$0.38R_u$+0.12F	$V_{right}=V_{left}$

Stage	Mass Factor K_M	Load Factor K_L	K_{LM} = K_M / K_L	Average K_{LM}
Elastic	0.5	0.64	0.78	0.72
Plastic	0.33	0.5	0.66	

Equivalent Mass = $M_e = K_{LM}$ x M = 0.72 x 0.00405 = 0.00292 $k\text{-}sec^2$ / in

Period of vibration = $t_n = 2 \times \pi \times \sqrt{(M_e / K)}$ =
2 x pi x SQRT (0.00292 / 227.16) = 0.023 sec

Damping ratio = ξ = 0 %

Damping constant = c = 2 x ξ x $\sqrt{(K \times M_e)}$ =
2 x 0 x SQRT (227.16 x 0.00292) / 100 = 0.000 k-sec/in

Blast load duration (For Side wall) = t_d = 0.0200 sec

Required Time step = min (t_d / 10 & t_n / 10) = 0.0020 sec

Time step considered for Numeriacl Integration = 0.0020

table above Support Reaction = V (t) = 0.385 R + 0.115 F

Blast Pressure on Side wall = P_r = 6.60 psi

Loading on Side wall = P_r H b = 6.6 x 9.84 x 12 x 12 / 1000 = 9.4 kips

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Input Data summary for Numerial Integration

Force (kips)	Time (sec)			
		K =	227.16	kip/in
		m = M_e =	0.00292	kips-sec^2 / in
9.4	0	c =	0.000	kips-sec/in
0	0.0200	Rt =	42.90	kips
0	0	Rc =	-42.90	kips

Numerical Integration solution (Side Wall)

time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.000	9.40	0.000	0.00	3219.2	0.00	1.08
0.002	8.46	0.006	5.66	2437.0	1.34	1.49
0.004	7.52	0.021	9.03	933.8	4.79	2.71
0.006	6.58	0.040	9.12	-845.6	9.05	4.24
0.008	5.64	0.055	5.89	-2374.9	12.57	5.49
0.010	4.70	0.062	0.32	-3201.5	14.05	5.95
0.012	3.76	0.056	-5.96	-3081.1	12.76	5.34
0.014	2.82	0.039	-11.09	-2049.2	8.80	3.71
0.016	1.88	0.014	-13.55	-411.0	3.08	1.40
0.018	0.94	-0.013	-12.62	1348.7	-3.00	-1.05
0.020	0.00	-0.035	-8.56	2709.5	-7.91	-3.05
0.022	0.00	-0.046	-2.27	3574.7	-10.44	-4.02
0.024	0.00	-0.043	4.68	3382.4	-9.88	-3.80
0.026	0.00	-0.028	10.25	2189.5	-6.39	-2.46
0.028	0.00	-0.004	12.79	348.8	-1.02	-0.39
0.030	0.00	0.021	11.55	-1595.0	4.66	1.79
0.032	0.00	0.039	6.88	-3067.0	8.96	3.45
0.034	0.00	0.047	0.19	-3631.7	10.60	4.08
0.036	0.00	0.040	-6.57	-3121.9	9.12	3.51
0.038	0.00	0.022	-11.38	-1688.6	4.93	1.90
0.040	0.00	-0.003	-12.82	244.2	-0.71	-0.27
0.042	0.00	-0.027	-10.47	2104.8	-6.15	-2.37
0.044	0.00	-0.043	-5.03	3342.8	-9.76	-3.76
0.046	0.00	-0.046	1.91	3591.8	-10.49	-4.04
0.048	0.00	-0.036	8.28	2778.2	-8.11	-3.12
0.050	0.00	-0.015	12.20	1142.8	-3.34	-1.28
0.052	0.00	0.011	12.51	-830.7	2.43	0.93
0.054	0.00	0.033	9.12	-2558.5	7.47	2.88
0.056	0.00	0.045	3.03	-3529.4	10.31	3.97
0.058	0.00	0.044	-3.95	-3456.1	10.09	3.89

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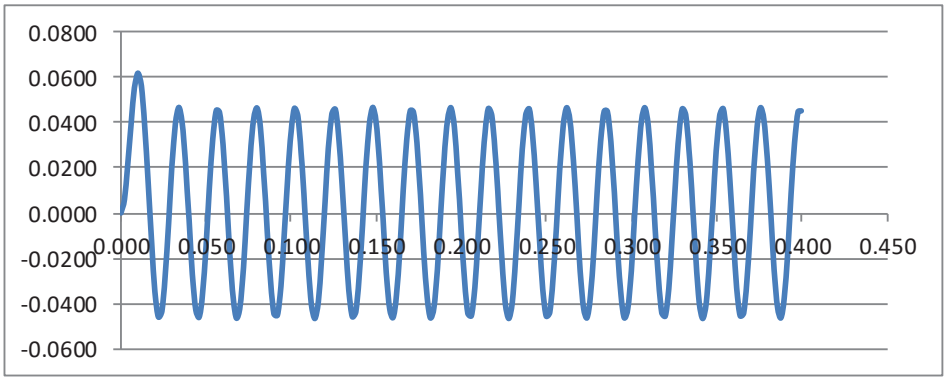
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time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.060	0.00	0.030	-9.77	-2360.4	6.89	2.65
0.062	0.00	0.007	-12.69	-566.4	1.65	0.64
0.064	0.00	-0.018	-11.87	1395.1	-4.07	-1.57
0.066	0.00	-0.038	-7.53	2944.0	-8.60	-3.31
0.068	0.00	-0.047	-0.96	3621.9	-10.58	-4.07
0.070	0.00	-0.041	5.89	3228.3	-9.43	-3.63
0.072	0.00	-0.024	11.00	1879.7	-5.49	-2.11
0.074	0.00	0.000	12.85	-25.0	0.07	0.03
0.076	0.00	0.025	10.90	-1922.3	5.61	2.16
0.078	0.00	0.042	5.73	-3250.9	9.49	3.65
0.080	0.00	0.047	-1.14	-3617.8	10.56	4.07
0.082	0.00	0.037	-7.67	-2914.4	8.51	3.28
0.084	0.00	0.017	-11.93	-1348.8	3.94	1.52
0.086	0.00	-0.008	-12.67	615.8	-1.80	-0.69
0.088	0.00	-0.031	-9.65	2398.2	-7.00	-2.70
0.090	0.00	-0.045	-3.78	3471.2	-10.14	-3.90
0.092	0.00	-0.045	3.21	3517.2	-10.27	-3.95
0.094	0.00	-0.032	9.25	2522.8	-7.37	-2.84
0.096	0.00	-0.010	12.55	782.0	-2.28	-0.88
0.098	0.00	0.015	12.14	-1190.1	3.48	1.34
0.100	0.00	0.036	8.14	-2810.2	8.21	3.16
0.102	0.00	0.046	1.73	-3598.9	10.51	4.05
0.104	0.00	0.043	-5.19	-3322.9	9.70	3.74
0.106	0.00	0.027	-10.58	-2063.9	6.03	2.32
0.108	0.00	0.002	-12.83	-194.3	0.57	0.22
0.110	0.00	-0.022	-11.30	1732.7	-5.06	-1.95
0.112	0.00	-0.040	-6.42	3147.2	-9.19	-3.54
0.114	0.00	-0.047	0.36	3630.6	-10.60	-4.08
0.116	0.00	-0.039	7.03	3039.9	-8.88	-3.42
0.118	0.00	-0.020	11.62	1550.0	-4.53	-1.74
0.120	0.00	0.005	12.77	-398.6	1.16	0.45
0.122	0.00	0.029	10.15	-2229.2	6.51	2.51
0.124	0.00	0.044	4.52	-3400.3	9.93	3.82
0.126	0.00	0.046	-2.45	-3565.5	10.41	4.01
0.128	0.00	0.034	-8.69	-2675.9	7.81	3.01
0.130	0.00	0.013	-12.36	-994.7	2.90	1.12
0.132	0.00	-0.013	-12.37	980.8	-2.86	-1.10
0.134	0.00	-0.034	-8.73	2666.1	-7.79	-3.00
0.136	0.00	-0.046	-2.50	3562.7	-10.40	-4.01
0.138	0.00	-0.044	4.47	3405.4	-9.94	-3.83

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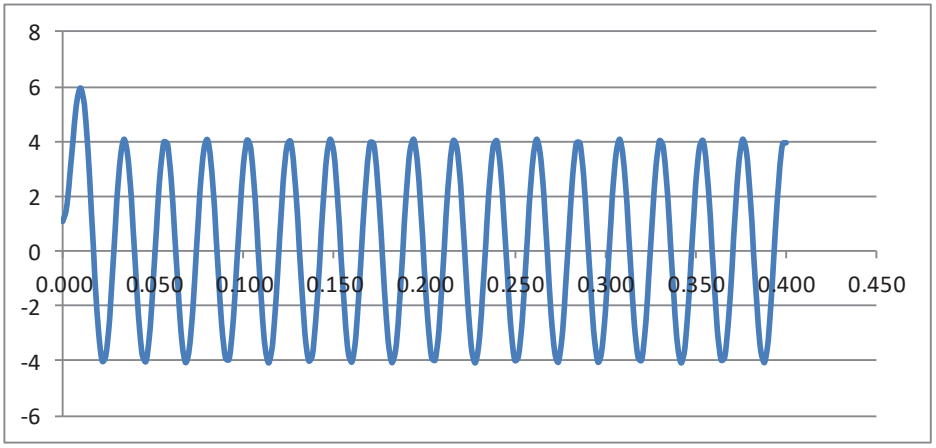
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Displacement versus time

max positive deflection y_{max} = 0.062 in @ t = 0.010 sec
max rebound deflection y_{reb} = -0.046 in @ t = 0.022 sec

Plastic Deformation = $y_p = y_m - y_e$ = 0.062 - 0.19 = 0.000 in
Rebound elastic deformation = $y_{reb} - y_p$ = 0.046 - 0 = 0.046 in



Reactions versus time

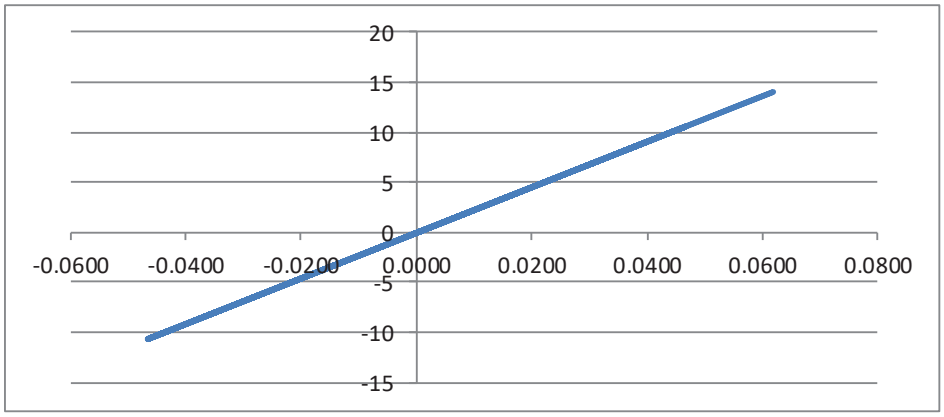
max dynamic reaction R_{dyn} = 5.95 kip/ft@ t= 0.010 sec
max rebound reaction R_{reb} = -4.02 kip/ft@ t= 0.022 sec

Support rotation check
$\theta = \arctan(2\ y_m / H)$ = Atan(2 x 0.062 / (9.84 x 12) = 0.06 deg
OK, rotation within 2 degree limit

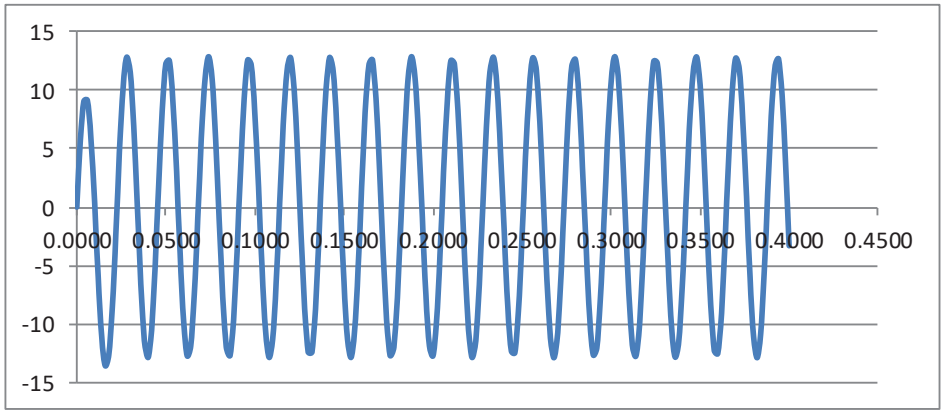
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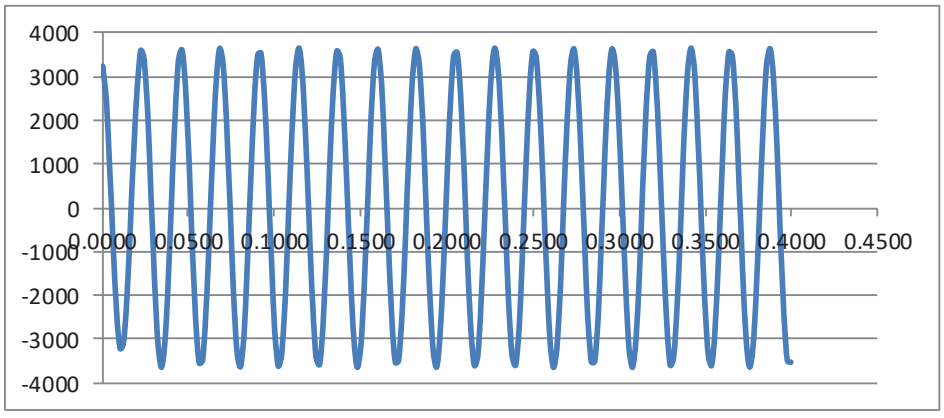
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Dynamic Resistance versus Displacement



Velocity versus Time



Acceleration versus Time

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9. DESIGN OF REAR WALL

Material properties

Grade of Concrete (f_{ck}) =	35.0	N/mm^2	5	ksi
Grade of Steel (f_y) =	415.0	N/mm^2	60	ksi
Con. tensile strength = 0.7 $\sqrt{f_{ck}}$	4.1	N/mm^2	1.6	ksi
Ec = 5000 $\sqrt{f_{ck}}$	29580.4	N/mm^2	4290	ksi
Es =	200000.0	N/mm^2	29008	ksi
Density of concrete (γ) =	25.0	kN/m^3	159	lb/ft^3
Modular ratio (n = Es/Ec) =	6.76		6.76	

Wall Section - Rear Wall

Geometrical properties, trial size

Height of wall	H =	3.00	m	9.84	ft
width of the section	b =	1.00	ft	12	in
overall depth	D =	300.00	mm	12	in
Clear Cover to reinforcement		30.00	mm	1.20	in
effective depth	d =	260	mm	10.41	in

Reinforcement in Rear wall

Vertical bar (outside the horizontal bar)

Dia	Spc.	Area	Dia	Area	Spc.	Area
mm	mm c/c	mm^2 / ft	in	in^2	in c/c	in^2 / ft
20	125	766	0.787	0.49	5	1.18

	SIF	DIF
Concrete	1	1.19
Steel	1.1	1.17

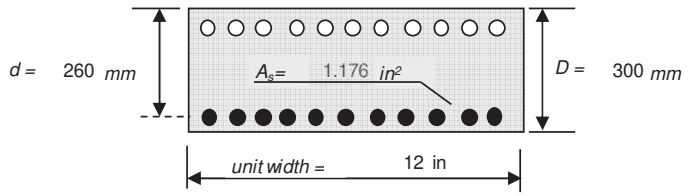
f_{dy} = (SIF) x (DIF) x f_y = (1.1) x (1.17) x 60 = 77.20 ksi

f_{dc} = (SIF) x (DIF) x f_{ck} = (1) x (1.19) x 5 = 5.95 ksi

A_s - Area of Reio. provided = 0.49(bar area) / 5 (spc.) x 12" = 1.176 in^2

Min. reo. Ratio = min [200 / fdy ; 3$\sqrt{}$fdc / fdy] = min[0.0026 ; 0.003] = 0.0026

ρ = Reo. Ratio provided = 1.176 / (12 x 10.4) = 0.0094 > As min, Ok



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Compute Bending Resistance:-

For dynamic Bending:

Depth of Neutral Axis = a = $A_s \times f_{dy} / (0.85 \times f'_{dc} \times b) =$
$1.18 \times 77.2 / (0.85 \times 5.95 \times 12) =$ 1.50 in

Plastic moment = $M_p = M_n =$ $A_s \times f_{dy} \times [d - a / 2] =$
$1.18 \times 77.2 \times [10.41 - 1.5 / 2] =$ 877 in-kip

$R_b = 8 M_p / L =$ $8 \times 877 / (9.84 \times 12) =$ 59.40 kips

Compute Shear Resistance:-

For dynamic Shear:

	SIF	DIF
Concrete	1	1

$f'_{dc} = (SIF) \times (DIF) \times f_{cu} =$ $(1) \times (1) \times 5 =$ 5.00 ksi

Nominal Shear strength = $V_n =$ $2 \times \sqrt{f'c} \times b \times d =$
$2 \times SQRT(5 \times 1000) \times 12 \times 10.41 / 1000 =$ 17.67 kips

Critical section for shear is at distance 'd' from face of support
$R_s = V_n L / (0.5 L - d) =$
$17.67 \times 9.84 \times 12 / ((0.5 \times 9.84 \times 12) - 10.41) =$ 42.90 kips

Compute SDOF Equivalent System:-

Because Rb > Rs, Shear Control

Governing Resistance, $R_u = \min(R_b \& R_s) =$ ***42.90*** kips

Gross Moment of Inertia = $I_g = b \times D^3 / 12 =$
$12 \times 12 \text{ ^3} / 12 =$ 1728 in^4

Cracked moment of inertia

$n \times A_s =$ $6.76 \times 1.176 =$ 7.95 in^2

$C = (-n A_s + \sqrt{} (n A_s (n A_s + 2 b d))) / b$
$((-7.95) + (SQRT (7.95 \times (7.95 + 2 \times 12 \times 10.41)))) / 12 =$ 3.11 in

Cracked Moment of inertia = $I_{cr} = (b \times C^3 / 3) + (n \times A_s \times (d - C)^2) =$
$(12 \times 3.11 \text{ ^3} / 3) + (7.95 \times (10.41 - 3.11) \text{ ^2}) =$ 544 in^4

Average moment of Inertia = $I_a = (Ig + Icr) / 2 =$ 1136 in^4

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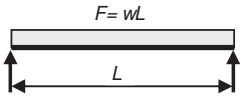
Effective Stiffness = K = 384 E I / 5 L^3 =
384 x 4290 x 1136 / (5 x (9.84 x 12) ^3) = 227.16 kip/in

Yield Deflecition = $y_e = R_u / K$ = 42.9 / 227.16 = 0.19 in

Acceleration due to gravity = g = 386 in/sec^2

Beam Mass = M = γ b D H / g =
(159 / 1000) x (12 / 12) x (12 / 12) x (9.84) / 386 = 0.00405 $k\text{-}sec^2$ / in

Ref Table 6.1



Strain Range	K_L	K_M^{lumped}	$K_M^{uniform}$	R_b	K	V_{left}	V_{right}
Elastic	0.64	n/a	0.50	$8M_{pc}/L$	$384EI/5L^3$	$0.39R+0.11F$	$V_{right}=V_{left}$
Plastic	0.50	n/a	0.33	$8M_{pc}/L$	0	$0.38R_u+0.12F$	$V_{right}=V_{left}$

Stage	Mass Factor K_M	Load Factor K_L	K_{LM} = K_M / K_L	Average K_{LM}
Elastic	0.5	0.64	0.78	0.72
Plastic	0.33	0.5	0.66	

Equivalent Mass = $M_e = K_{LM}$ x M = 0.72 x 0.00405 = 0.00292 $k\text{-}sec^2$ / in

Period of vibration = $t_n = 2 \times \pi \times \sqrt{(M_e / K)}$ =
2 x pi x SQRT (0.00292 / 227.16) = 0.023 sec

Damping ratio = ξ = 0 %

Damping constant = c = 2 x ξ x $\sqrt{(K \times M_e)}$ =
2 x 0 x SQRT (227.16 x 0.00292) / 100 = 0.000 k-sec/in

Blast load duration (For Rear wall) = t_d = 0.0270 sec

Required Time step = min (t_d / 10 & t_n / 10) = 0.0023 sec

Time step considered for Numeriacl Integration = 0.0020

table above Support Reaction = V (t) = 0.385 R + 0.115 F

Blast Pressure on Rear wall = P_r = 4.90 psi

Loading on Rear wall = P_r H b = 4.9 x 9.84 x 12 x 12 / 1000 = 6.9 kips

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Input Data summary for Numerial Integration

Force (kips)	Time (sec)
0	0
0	0.088
6.9	0.095
0	0.115

K = 227.16 kip/in
$m = M_e$ = 0.00292 kips-sec^2 / in
c = 0.00 kips-sec/in
Rt = 42.90 kips
Rc = -42.90 kips

Numerical Integration solution (Rear Wall)

time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.120	0.00	-0.003	7.92	206.4	-0.60	-0.23
0.122	0.00	0.013	7.13	-995.2	2.91	1.12
0.124	0.00	0.024	4.23	-1902.4	5.56	2.14
0.126	0.00	0.029	0.08	-2246.9	6.56	2.53
0.128	0.00	0.025	-4.09	-1926.6	5.63	2.17
0.130	0.00	0.013	-7.05	-1036.3	3.03	1.17
0.132	0.00	-0.002	-7.93	160.5	-0.47	-0.18
0.134	0.00	-0.017	-6.46	1309.8	-3.82	-1.47
0.136	0.00	-0.027	-3.08	2071.7	-6.05	-2.33
0.138	0.00	-0.029	1.21	2220.6	-6.48	-2.50
0.140	0.00	-0.022	5.15	1712.7	-5.00	-1.93
0.142	0.00	-0.009	7.56	698.0	-2.04	-0.78
0.144	0.00	0.007	7.73	-523.1	1.53	0.59
0.146	0.00	0.020	5.62	-1589.5	4.64	1.79
0.148	0.00	0.028	1.84	-2185.7	6.38	2.46
0.150	0.00	0.027	-2.48	-2135.2	6.23	2.40
0.152	0.00	0.019	-6.06	-1453.1	4.24	1.63
0.154	0.00	0.004	-7.86	-341.1	1.00	0.38
0.156	0.00	-0.011	-7.33	871.8	-2.55	-0.98
0.158	0.00	-0.023	-4.63	1826.8	-5.33	-2.05
0.160	0.00	-0.029	-0.56	2241.4	-6.54	-2.52
0.162	0.00	-0.026	3.67	1992.9	-5.82	-2.24
0.164	0.00	-0.015	6.82	1154.8	-3.37	-1.30
0.166	0.00	0.000	7.95	-24.9	0.07	0.03
0.168	0.00	0.015	6.73	-1197.2	3.50	1.35
0.170	0.00	0.026	3.52	-2015.4	5.88	2.27
0.172	0.00	0.029	-0.74	-2237.3	6.53	2.52
0.174	0.00	0.023	-4.77	-1797.4	5.25	2.02
0.176	0.00	0.011	-7.39	-825.7	2.41	0.93
0.178	0.00	-0.005	-7.83	390.2	-1.14	-0.44

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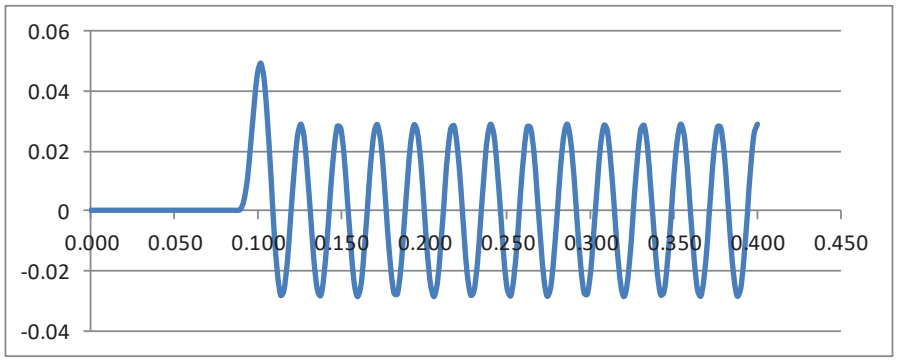
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time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.180	0.00	-0.019	-5.95	1490.7	-4.35	-1.68
0.182	0.00	-0.028	-2.31	2150.2	-6.28	-2.42
0.184	0.00	-0.028	2.02	2173.6	-6.35	-2.44
0.186	0.00	-0.020	5.74	1553.9	-4.54	-1.75
0.188	0.00	-0.006	7.77	474.6	-1.39	-0.53
0.190	0.00	0.010	7.50	-745.2	2.18	0.84
0.192	0.00	0.022	5.01	-1744.5	5.09	1.96
0.194	0.00	0.029	1.04	-2227.7	6.50	2.50
0.196	0.00	0.026	-3.24	-2051.9	5.99	2.31
0.198	0.00	0.016	-6.56	-1269.1	3.71	1.43
0.200	0.00	0.001	-7.94	-110.8	0.32	0.12
0.202	0.00	-0.014	-6.97	1080.2	-3.15	-1.21
0.204	0.00	-0.025	-3.94	1951.7	-5.70	-2.19
0.206	0.00	-0.029	0.26	2245.8	-6.56	-2.52
0.208	0.00	-0.024	4.38	1875.5	-5.48	-2.11
0.210	0.00	-0.012	7.20	950.4	-2.78	-1.07
0.212	0.00	0.003	7.90	-255.9	0.75	0.29
0.214	0.00	0.018	6.26	-1386.5	4.05	1.56
0.216	0.00	0.027	2.76	-2106.9	6.15	2.37
0.218	0.00	0.028	-1.55	-2204.0	6.44	2.48
0.220	0.00	0.021	-5.40	-1649.1	4.82	1.85
0.222	0.00	0.008	-7.66	-606.3	1.77	0.68
0.224	0.00	-0.008	-7.65	615.8	-1.80	-0.69
0.226	0.00	-0.021	-5.37	1655.8	-4.83	-1.86
0.228	0.00	-0.028	-1.51	2205.9	-6.44	-2.48
0.230	0.00	-0.027	2.80	2103.4	-6.14	-2.36
0.232	0.00	-0.018	6.28	1378.7	-4.03	-1.55
0.234	0.00	-0.003	7.90	246.1	-0.72	-0.28
0.236	0.00	0.012	7.19	-959.3	2.80	1.08
0.238	0.00	0.024	4.35	-1880.9	5.49	2.11
0.240	0.00	0.029	0.22	-2246.1	6.56	2.53
0.242	0.00	0.025	-3.97	-1946.8	5.68	2.19
0.244	0.00	0.014	-6.99	-1071.5	3.13	1.20
0.246	0.00	-0.002	-7.94	120.7	-0.35	-0.14
0.248	0.00	-0.016	-6.54	1277.2	-3.73	-1.44
0.250	0.00	-0.026	-3.21	2055.9	-6.00	-2.31
0.252	0.00	-0.029	1.07	2226.4	-6.50	-2.50
0.254	0.00	-0.022	5.04	1738.2	-5.08	-1.95

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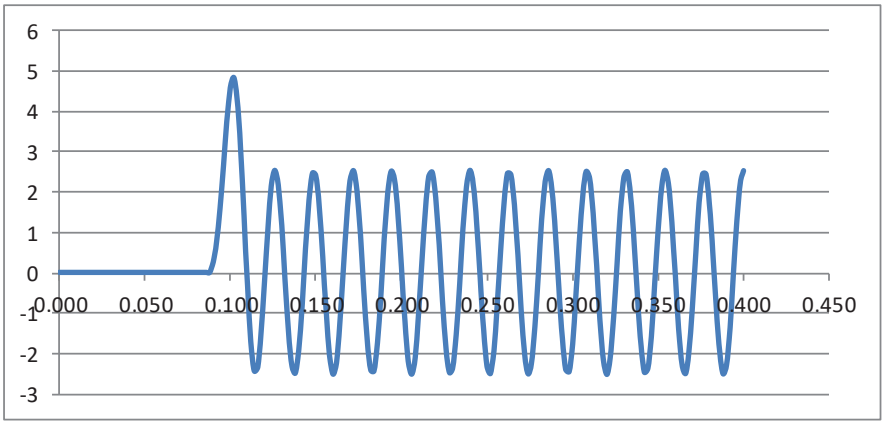
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Displacement versus time

max positive deflection y_{max} = 0.049 in @ t = 0.102 sec
max rebound deflection y_{reb} = -0.028 in @ t = 0.114 sec

Plastic Deformation = $y_p = y_m - y_e$ = 0.049 - 0.19 = 0.000 in
Rebound elastic deformation = $y_{reb} - y_p$ = -0.028 - 0 = -0.028 in



Reactions versus time

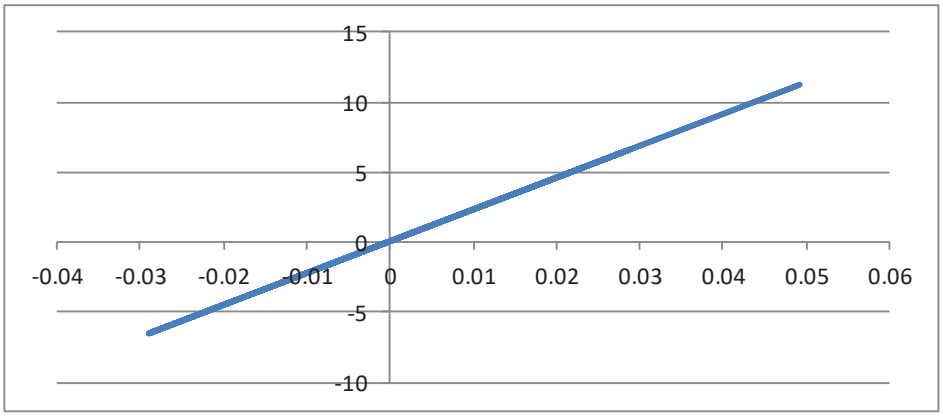
max dynamic reaction R_{dyn} = 4.824 kip/ft@ t= 0.102 sec
max rebound reaction R_{reb} = -2.442 kip/ft@ t= 0.114 sec

Support rotation check
$\theta = \arctan(2\, y_m / H)$ = Atan(2 x 0.049 / (9.84 x 12) = 0.05 deg
OK, rotation within 2 degree limit

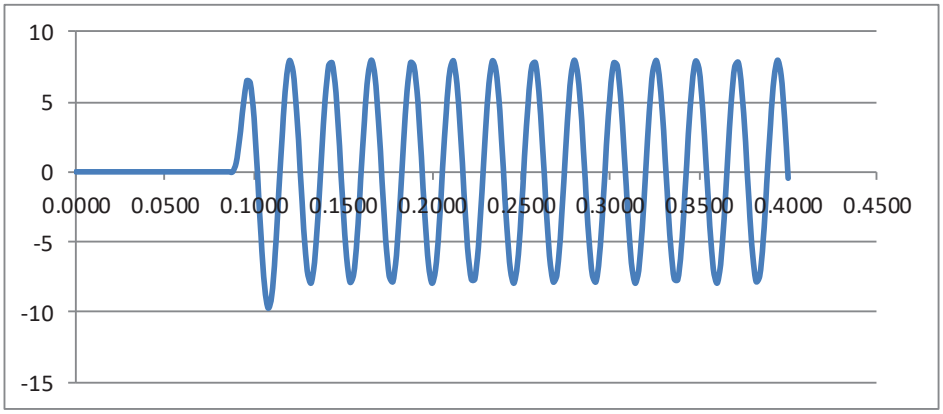
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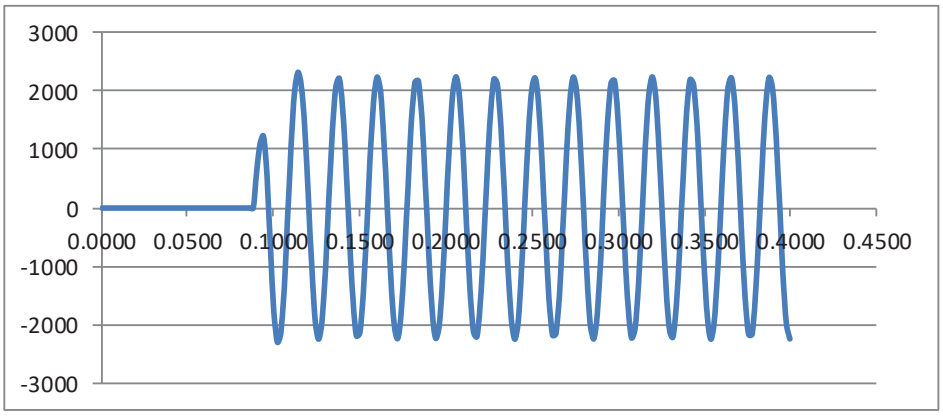
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Dynamic Resistance versus Displacement



Velocity versus Time



Acceleration versus Time

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10. SIDE WALL (IN PLANE)

Material properties

Grade of Concrete (f_{ck}) =	35 N/mm^2	5 ksi
Grade of Steel (f_y) =	415 N/mm^2	60 ksi
Con. tensile strength = 0.7 $\sqrt{f_{ck}}$	4 N/mm^2	1.6 ksi
Ec = 5000 $\sqrt{f_{ck}}$	29580 N/mm^2	4290 ksi
Es =	200000 N/mm^2	29008 ksi
Density of concrete (g) =	25 kN/m^3	159 lb / ft^3
Modular ratio (n = Es/Ec) =	6.76	6.76

Side wall - (in-plane-bending)

Height of wall (from top of footing / pile cap)	3.0 m	9.84 ft
Length of wall = B	36 m	118.11 ft
Wall thickness	300 mm	12 in
Depth of building = L =	40 m	131.23 ft
Thickness of roof slab =	20 mm	1 in
Wall height from C/L of base slab =	3.0 m	9.84 ft

Load Case A - Use the reactions from roof diaphragm analysis

Load Case B - neglect this case because it will not control

Side wall should match front wall design

Wall Reinforcement

Vertical bar (outside the horizontal bar)

Dia mm	Spc. mm c/c	Area mm^2 / ft	Dia in	Area in^2	Spc. in c/c	Area in^2 / ft
20	125	766.05	0.787	0.49	5	1.18

Clear cover to vertical reinforcement = 30.0 mm 1 in

Horizontal Bar (inside vertical bar)

Dia mm	Spc. mm c/c	Area mm^2 / ft	Dia in	Area in^2	Spc. in c/c	Area in^2 / ft
16	150	408.56	0.630	0.31	5.906	0.630

Compute Bending Resistance

For dynamic bending

	SIF	DIF	SIF x DIF	fd	
Concrete	1	1.19	1.19	5.95	= f'_{dc} (ksi)
Steel	1.1	1.17	1.287	77.2	= f'_{dy} (ksi)

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For Bending, assume 15 bars at the corner provide the tension component for resisting in-plane bending.
Because of this approximation, the in-plane response will be limited to the elastic range

A_s = 0.49(area) x 15(nos) = 7.35 in^2

b_w = (wall thickness) = 12 in

Effective depth d = B - wall thk. + clear cover + bar dia /2 =
(118.11 x 12) - 12 + 1 + 0.787 / 2 = 1407 in

ρ = ratio of reinforcement provided = 0.0004
Min. reo. Ratio = min [200 / f_{dy} ; 3$\sqrt{f_{dc}}$ / f_{dy}] min[0.0026 ; 0.003] = 0.0026 < As min
Keep the response limited to elastic range

b_f = minimum of following values =
a) 1/12 x Span of beam = 1 / 12 x 9.84 x 12 = 9.84 in
b) 6 x wall thickness = 6 x 12 = 72 in
c)1/2 x clear distance between next web 1 / 2 x 131.23 x 12 = 787.38 in

Use beam flange width = 9.84 in say 24 in

a = A_s x f_{dy} / (0.85 x f'_{dc} x b) =
7.35 x 77.2 / (0.85 x 5.95 x 24) = 4.67 in

M_p = A_s x f_{dy} x [d - a / 2] = 7.35 x 77.2 x [1407 - 4.67 / 2] = 797035 in-k

R_b = M_p / L = M_p / H = 797035 / (9.84 x 12) = 6750 kips

Compute Shear Resistance from Shear Friction Criteria

	SIF	DIF	SIF x DIF	fd	
Steel	1.1	1.1	1.21	72.6	= f_{dy} (ksi)

Friction coefficient μ = 0.6 (λ) = (not intentionally roughened) 0.6

A_{vf} = Vertical reo Area x B x 2 sides 0.49 / 5 x (118.11 x 12) x 2 = 277.79 in^2

R_s = V_n = A_{vf} x f_{dy} x m = 277.79 x 72.6 x 0.6 = 12101 kips

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Compute SDOF Equivalent System

By inspection Rb < Rs, Bending controls

Ru = Min (6750 , 12101) = 6750 kips

Because of analysis approximation and because minimum reinforcement was not provided use Ductility ratio = μ_a = 1

Because the shear wall is a deep and relatively short beam, the stiffness must include shear deformations. Compute the total deflection for an arbitary load of P = 1000 kips.

For moment if inertia calculation, to approximate the effect of wall cracking due to out-of-plane loads, use half the wall thickness

Thickness of wall to be consideed = 12 / 2 = 6 in

Width of front / rear wall acting as chord = 4 b + t = 4 x 12 + 12 = 60 in

include chord reinforcing = n As = 6.76 x 7.35 = 49.69 in^2

Clear dist. between wall = B - 2 wall thk.= 118.11 x 12 - 2 x 12 = 1393 in

C.G. of chord from C/L of wall = (118.11 x 12 / 2) - (12 / 2) = 703 in

$I = S\ b\ h^3 / 12 + S\ A\ d^2$ =

(2 x 60 x 12 ^3 / 12) + (6 x 1393 ^3 / 12) + ((2 x (60 x 12 + 49.69)) x 703 ^2) = 2.11E+09 in^4

Flexural deflection = $f = P\ H^3 / 3\ E\ I$ =

1000 x (9.84 x 12) ^3 / (3 x 4290 x 2112315459) = 0.000 in

Poisson's Ratio = ν = 0.2

Shear Modulus = $G = E_c / 2 \times (1 + \nu)$ = 4290 / (2 x (1 + 0.2)) = 1787.500 ksi

Shear Deflection = $s = 1.2\ H\ P / AG$ =

1.2 x 9.84 x 12 x 1000 / (12 x 118.11 x 12 x 1787.5) = 0.0047 in

Beam Stiffness = $K = P / (f + s)$ = 1000 / (0 + 0.0047) = 212766 k/in

yiels deflection = $y_e = R_u / K$ = 6750 / 212766 = 0.032 in

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Tributary front & rear wall weight

W_1 = 12 / 12 x 131.23 / 2 x 9.84 / 2 x 2 (nos.) x 159 / 1000 = 102.7 kip

Tributary roof weight

W_2 = 131.23 / 2 x 118.11 x 1 / 12 x 159 /1000 = 102.7 kip

Shear Wall weight

W_3 = 118.11 x 9.84 / 2 x 12 / 12 x 159 /1000 = 92.4 kip

Include 20 % of the front wall and roof with shear wall mass.

Acceleration due to gravity = g = 9.810 m/sec^2 386 in/sec^2

Beam Mass - M = [W3 + 0.2 x (W2 + W1)] / 386 = 0.346 k-sec^2 / in

Because of the expected response, use elastic values to compute K_{LM}

Stage	Mass Factor K_M	Load Factor K_L	K_{LM} = K_M / K_L
Elastic	1.000	1.000	1.00

Equivalent mass = $M_e = K_{LM}$ x M = 1 x 0.35 = 0.346 k-sec^2/in

Period of vibration = $t_n = 2 \times p \times \sqrt{(M_e / K)}$ =

2 x pi x SQRT (0.346 / 212766) = 0.008 sec

Damping ratio = ξ = 0 %

Damping constant = $c = 2 \times \xi \times \sqrt{(K \times M_e)}$ =

2 x 0 x SQRT (212766 x 0.346) / 100 = 0.000 k-sec/in

Time step considered for Numeriacl Integration = 0.0020

Support Reaction = V (t) = 1.000 R + 0.000 F

Input Data summary for Numerial Integration

K =	212766 kip/in
m = M_e =	0.346 kips-sec^2 / in
c =	0.000 kips-sec/in
Rt =	6750 kips
Rc =	-6750 kips

For in-plane bending of Side Wall, dynamic reactions from Roof in-plane bending will be the force acting on Side wall

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Numerical Integration solution (Side Wall - in-plane-loading)

time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.000	49.06	0.000	0.00	141.8	0.00	0.00
0.002	74.12	0.000	0.21	69.5	50.08	50.08
0.004	152.26	0.001	0.27	-15.2	157.53	157.53
0.006	270.68	0.001	0.26	9.2	267.50	267.50
0.008	413.14	0.002	0.34	66.9	390.00	390.00
0.010	561.26	0.003	0.42	19.5	554.52	554.52
0.012	695.98	0.003	0.35	-89.4	726.90	726.90
0.014	799.20	0.004	0.16	-106.9	836.18	836.18
0.016	855.34	0.004	0.01	-34.5	867.26	867.26
0.018	852.81	0.004	-0.04	-22.2	860.48	860.48
0.020	785.09	0.004	-0.17	-104.8	821.36	821.36
0.022	654.25	0.003	-0.41	-134.0	700.61	700.61
0.024	468.37	0.002	-0.58	-42.2	482.98	482.98
0.026	240.88	0.001	-0.59	37.9	227.76	227.76
0.028	-8.12	0.000	-0.54	7.8	-10.82	-10.82
0.030	-256.51	-0.001	-0.57	-34.7	-244.52	-244.52
0.032	-482.19	-0.002	-0.57	29.9	-492.53	-492.53
0.034	-665.13	-0.003	-0.41	129.9	-710.08	-710.08
0.036	-789.25	-0.004	-0.16	123.9	-832.11	-832.11
0.038	-843.83	-0.004	0.01	44.3	-859.15	-859.15
0.040	-824.41	-0.004	0.09	39.1	-837.95	-837.95
0.042	-733.14	-0.004	0.24	113.0	-772.24	-772.24
0.044	-578.38	-0.003	0.48	119.9	-619.86	-619.86
0.046	-373.92	-0.002	0.62	19.5	-380.67	-380.67
0.048	-137.65	-0.001	0.58	-49.7	-120.46	-120.46
0.050	109.98	0.001	0.53	-8.9	113.06	113.06
0.052	347.75	0.002	0.54	27.2	338.34	338.34
0.054	555.36	0.003	0.53	-46.0	571.26	571.26
0.056	715.10	0.004	0.34	-137.1	762.53	762.53
0.058	813.25	0.004	0.09	-115.3	853.14	853.14
0.060	841.26	0.004	-0.06	-36.2	853.77	853.77
0.062	796.44	0.004	-0.14	-43.2	811.40	811.40
0.064	682.30	0.003	-0.30	-117.0	722.77	722.77
0.066	508.36	0.003	-0.53	-109.2	546.16	546.16
0.068	289.42	0.001	-0.64	-3.2	290.52	290.52
0.070	44.44	0.000	-0.59	55.0	25.39	25.39
0.072	-205.15	-0.001	-0.52	7.8	-207.84	-207.84
0.074	-437.29	-0.002	-0.53	-17.3	-431.31	-431.31
0.076	-631.30	-0.003	-0.49	66.0	-654.12	-654.12
0.078	-769.83	-0.004	-0.27	147.9	-820.99	-820.99

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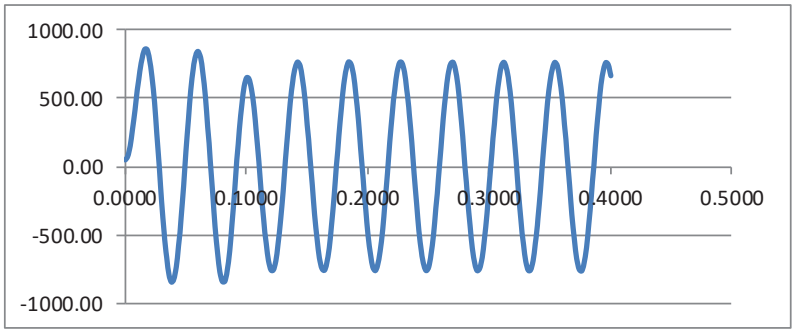
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time (sec)	force (kips)	y (in)	v (in / sec)	a (in / sec^2)	resistance (kips)	reaction (kips)
0.080	-840.50	-0.004	-0.01	110.9	-878.88	-878.88
0.082	-837.05	-0.004	0.13	32.4	-848.26	-848.26
0.084	-759.94	-0.004	0.21	48.4	-776.67	-776.67
0.086	-616.23	-0.003	0.38	116.5	-656.53	-656.53
0.088	-418.89	-0.002	0.58	91.3	-450.49	-450.49
0.090	-190.55	-0.001	0.65	-29.6	-180.31	-180.31
0.092	48.11	0.000	0.54	-77.7	75.01	75.01
0.094	270.68	0.001	0.44	-23.2	278.72	278.72
0.096	455.77	0.002	0.41	-5.0	457.51	457.51
0.098	585.72	0.003	0.31	-91.1	617.23	617.23
0.100	648.11	0.003	0.07	-156.7	702.34	702.34
0.102	639.74	0.003	-0.19	-94.0	672.27	672.27
0.104	563.43	0.003	-0.29	-6.6	565.70	565.70
0.106	427.75	0.002	-0.32	-29.4	437.91	437.91
0.108	246.30	0.001	-0.45	-94.8	279.09	279.09
0.110	36.55	0.000	-0.59	-52.9	54.84	54.84
0.112	-181.56	-0.001	-0.58	64.1	-203.75	-203.75
0.114	-387.28	-0.002	-0.42	94.6	-420.02	-420.02
0.116	-561.67	-0.003	-0.30	24.3	-570.07	-570.07
0.118	-688.45	-0.003	-0.27	9.2	-691.62	-691.62
0.120	-755.42	-0.004	-0.16	100.7	-790.25	-790.25
0.122	-756.42	-0.004	0.09	151.8	-808.94	-808.94
0.124	-691.05	-0.003	0.32	74.2	-716.71	-716.71
0.126	-564.83	-0.003	0.38	-8.1	-562.02	-562.02
0.128	-388.76	-0.002	0.40	26.0	-397.74	-397.74
0.130	-178.35	-0.001	0.51	85.1	-207.80	-207.80
0.132	47.75	0.000	0.62	28.0	38.08	38.08
0.134	269.50	0.001	0.57	-86.9	299.57	299.57
0.136	467.29	0.002	0.38	-99.3	501.64	501.64
0.138	623.72	0.003	0.26	-23.2	631.76	631.76
0.140	725.18	0.003	0.21	-19.3	731.87	731.87
0.142	763.05	0.004	0.08	-112.1	801.82	801.82
0.144	734.32	0.004	-0.17	-145.8	784.77	784.77
0.146	641.83	0.003	-0.38	-55.9	661.17	661.17
0.148	493.88	0.002	-0.41	17.8	487.71	487.71
0.150	303.47	0.001	-0.42	-26.6	312.67	312.67
0.152	87.10	0.001	-0.53	-77.8	114.03	114.03
0.154	-136.59	-0.001	-0.61	-8.3	-133.72	-133.72
0.156	-348.43	-0.002	-0.52	100.0	-383.03	-383.03
0.158	-530.24	-0.003	-0.33	95.4	-563.25	-563.25
0.160	-666.37	-0.003	-0.21	18.0	-672.60	-672.60
0.162	-744.99	-0.004	-0.17	27.1	-754.36	-754.36

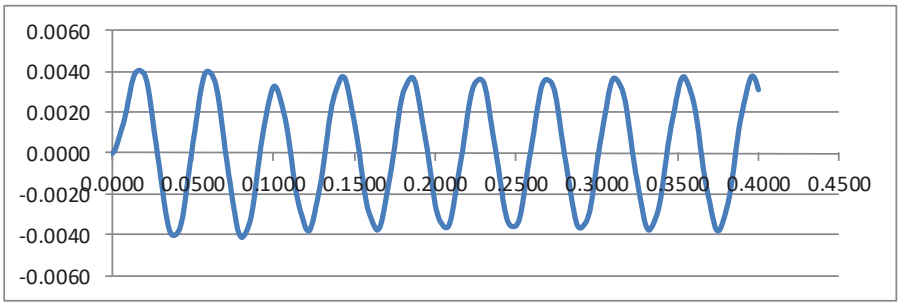
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Side wall - in-plane loading



Displacement versus time

max positive deflection y_{max}= 0.004 in @ t = 0.016 sec
max rebound deflection y_{reb}= -0.004 m @ t = 0.038 sec

Plastic Deformation = $y_p = y_m - y_e$ = 0.004 - 0.032 = 0.000 in
Rebound elastic deformation = $y_{reb} - y_p$ = -0.004 - 0 = -0.004 in

Ductility Ratio = $\mu_d = y_m / y_e$ = 0.004 / 0.032 = 0.13 < 1, Ok

the response of the member is ***elastic***
ymax < ye, Ok!
--

Support rotation check
θ = arctan(2 y_m / H) = Atan(2 x 0.004 / (9.84 x 12) = 0.004 deg
OK, rotation within 2 degree limit

Δ_d = Computed deformation (Ductility ratio OR support rotation)
Δ_a = Allowable deformation (Ductility ratio OR support rotation)
i = in-plane deformations
o = out-of-plane deformations

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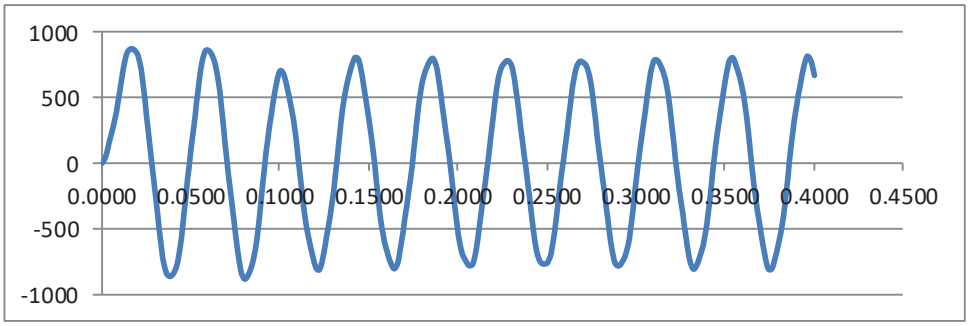
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For side wall

Δ_{di} =	0.13	Δ_{ai} =	1	Ductility ratio
Δ_{do} =	0.06	Δ_{ao} =	1	Rotation

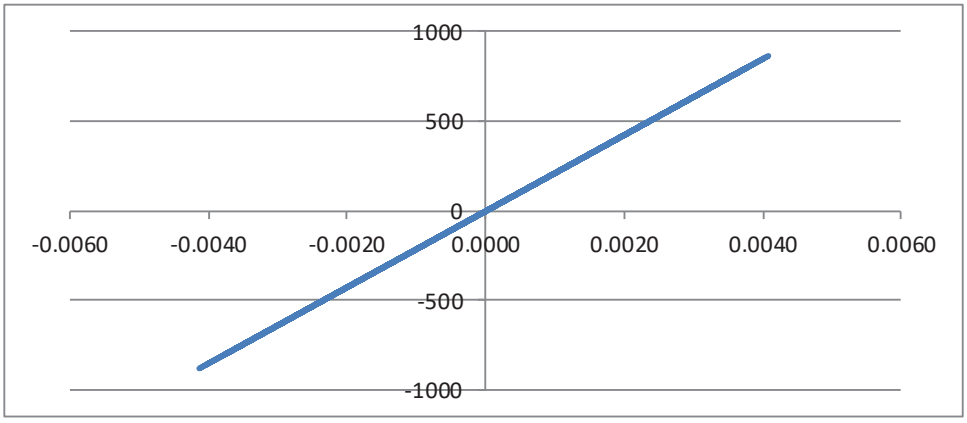
$[\Delta_d / \Delta_a]_i^2 + [\Delta_d / \Delta_a]_i^2 =$ [0.13 / 1] ^2 + [0.06 / 1] ^2 = 0.021 < 1, OK



Reaction versus time

max dynamic reaction R_{dyn}= 867.26 kip @ t= 0.02 sec

max rebound reaction R_{reb}= -859.15 kip @ t= 0.038 sec

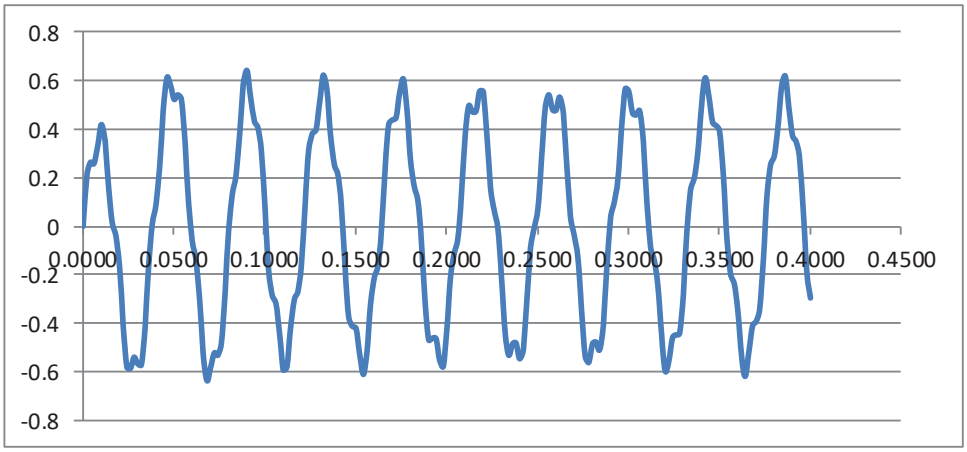


Dynamic Resistance versus Displacement

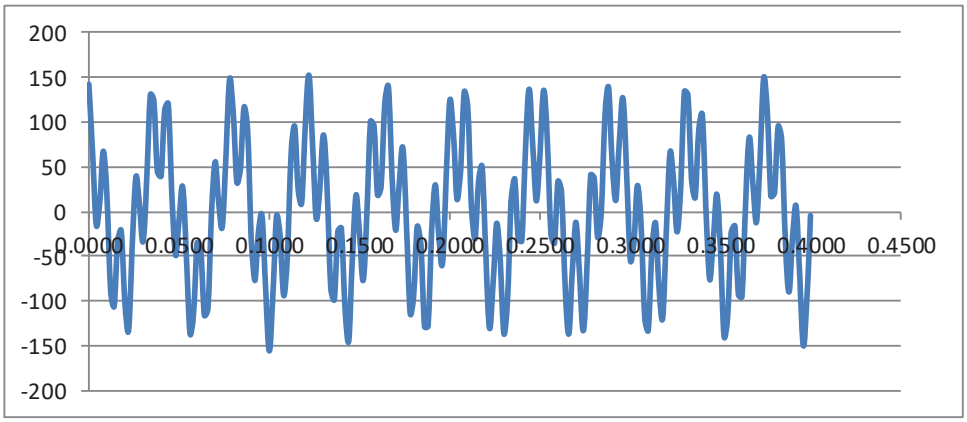
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Velocity versus Time



Acceleration versus Time

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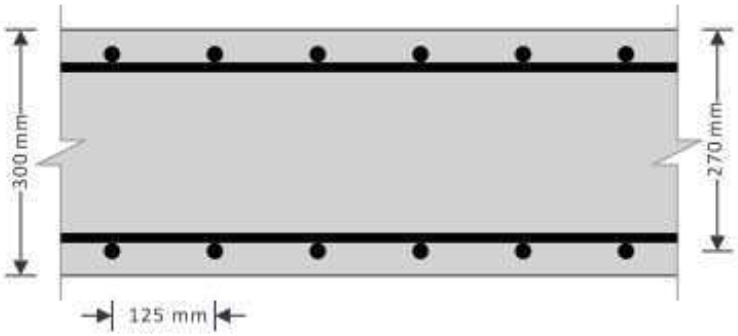
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11. DESIGN OF RC WALL (STRENGTH CHECK)

RC WALL DESIGN (ACI318-14)

TEDDS calculation version 1.2.02 -



Geometry of wall

Depth of wall	h = **300** mm
Clear cover to reinforcement (both sides)	c_c = **20** mm
Unsupported height of wall	l_u = **3000** mm
Effective height factor	k = **1.00**

Reinforcement of wall

Numbers of reinforcement layers	N_l = **2**
Spacing of vertical steel	s_v = **125** mm
Diameter of vertical steel bar	D_{ver} = **20** mm
Spacing of horizontal steel	s_h = **150** mm
Diameter of horizontal bar	D_{hor} = **12** mm
Specified yield strength of reinforcement	f_y = **414** MPa
Specified compressive strength of concrete	f'_c = **35** MPa
Modulus of elasticity of bar reinforcement	E_s = **200000** MPa
Modulus of elasticity of concrete	$E_c = 4700 \times \sqrt{(f'_c \times 1\ MPa)}$ = **27806** MPa
Ultimate design strain	ε_c = **0.003** mm/mm
Concrete modification factor	λ = **1.00**

Check for minimum area of vertical steel of double layer reinforcement wall cl. 11.6.1

Gross area of wall per running meter length	$A_g = h \times 1000\ mm$ = **300000** mm²
Numbers of vertical bars per running meter length	$N_v = 1000\ mm / s_v$ = **8.0**
Area of vertical steel per running meter length	$A_{st_v} = 2 \times N_v \times (\pi \times D_{ver}^2) / 4$ = **5027** mm²
Minimum area of vertical steel required	$A_{st_v_min}$ = **450** mm²

PASS - vertical steel provided is greater than minimum vertical steel required

Check for minimum area of horizontal steel of double layer reinforcement wall cl. 11.6.1

Gross area of wall per running meter length	$A_g = h \times 1000\ mm$ = **300000** mm²
Number of horizontal bar per running meter height	$N_h = 1000\ mm / s_h$ = **6.667**
Area of horizontal steel per running meter height	$A_{st_h} = 2 \times N_h \times (\pi \times D_{hor}^2) / 4$ = **1508** mm²

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Minimum area of horizontal steel required	$A_{st_h_min}$ = **600** mm²
	PASS - horizontal steel provided is greater than minimum horizontal steel required
Braced wall slenderness check 6.2.5	
Maximum slenderness ratio limit	s_{r_max} = 100
Permissible slenderness ratio	s_{r_perm} = min(34 + 12 × (M_{1_act} / M_{2_act}), 40) = **40.0**
Slenderness check for braced wall	
Radius of gyration	r_{min} = 0.3 × h = **90** mm
Actual slenderness ratio	s_{r_act} = k × l_u / r_{min} = **33.33**
	Wall slenderness limit OK, wall is braced short wall
Design loads and moments for wall subjected to shear, axial load and bending	
Ultimate axial force per running meter	P_{u_act} = **20.00** kN/m
Ultimate large end moment per running meter	M_{2_act} = **240.00** kNm/m
Ultimate small end moment per running meter	M_{1_act} = **125.00** kNm/m
Ultimate shear force per running meter	V_{u_act} = **180.00** kN/m
Ratio of DL moment to total moment	β_d = **0.650**
Axial load capacity of double layer reinforcement wall subjected to bending	
c/d_t ratio	r = **0.140**
Effective cover to reinforcement	d' = c_c + (D_{ver} / 2) = **30.0** mm
Depth of tension steel	d_t = h - d' = **270.0** mm
Depth of NA from extreme compression face	c = r × d_t = **37.8** mm
Factor of depth of comp. stress block	β_1 = **0.800**
Depth of equivalent rectangular stress block	a = min((β_1 × c), h) = **30.2** mm
Strain in 'compression' reinforcement	ε'_s = ε_c × (1 - d' / c) = **0.000619**
f'$_s$ Stress in 'compression' reinforcement	f'_s = min(E_s × ε'_s, f_y) = **123.8** MPa
Strain in 'tension' reinforcement	ε_s = ε_c × (1 - d_t / c) = **-0.018429**
f_s Stress in 'tension' reinforcement	f_s = max(E_s × ε_s, -f_y) = **-414.0** MPa
Compression force in concrete	C_c = 0.85 × f'$_c$ × a × 1000 mm /1 m= **899.64** kN/m
Area of vertical tension steel per running meter	A_s = A_{st_v} / 2 = **2513** mm²
Area of vertical comp. steel per running meter	A'$_s$ = A_{st_v} / 2 = **2513** mm²
Force in 'compression' steel	C_s = A'$_s$ × f_s / 1 m = **311.17** kN/m
Force in 'tension' steel	T_s = A_s × f_s / 1 m = **-1040.50** kN/m
Nominal axial load strength	P_n = C_c + C_s + T_s = **170.3** kN/m
Strength reduction factor	ϕ = 0.9 = 0.900
Ultimate axial load carrying capacity of wall	P_u = ϕ × P_n = **153.28** kN/m
	PASS - wall is safe in axial loading
Bending capacity of wall	
Centroid of wall	y = h × 0.5 = **150.0** mm
Nominal moment strength	M_n = C_c × (y - 0.5 × a) + C_s × (y - d') - T_s × (d_t – y) = **283.54** kNm/m
Ultimate moment strength capacity of wall	M_r = ϕ × M_n = **255.19** kNm/m
	PASS - wall is safe for bending
Check for shear capacity of wall subjected to shear, axial load and bending – cl. 22.5	
Strength reduction factor	ϕ_v = **0.75**

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Effective cover to reinforcement	$d' = c_c + (D_{ver} / 2) = \mathbf{30}$ mm
Depth of tension steel	$d_t = h - d' = \mathbf{270}$ mm
Factored moment for axial compression	$M_n = M_{2_act} - (P_{u_act} \times ((4 \times h) - d_t) / 8) = \mathbf{237.68}$ kNm/m
Shear force resisting capacity of wall	$V_{c1} = (0.16 \times \lambda \times \sqrt{(f'_c \times 1MPa)} \times d_t) + (17\ N/mm^2 \times A_n / 1\ m \times \min(1, (V_{u_act} \times d_t / M_n)))$
Shear force resisting capacity of wall	$V_{c1} = \mathbf{264.31}$ kN/m
Max. shear force resisting capacity of wall	$V_{max} = 0.29 \times \lambda \times \sqrt{(f'_c)} \times h \times \sqrt{(1\ MPa + (0.29 \times P_{u_act} \times 1\ m / A_g))}$
Max. shear force resisting capacity of wall	$V_{max} = \mathbf{519.65}$ kN/m
Shear force resisting capacity of wall	$\phi V_c = \phi_v \times \min(V_{c1}, V_{max}) = 198.23$ kN/m

PASS - shear force resisting capacity of wall is greater than shear force acting on wall

Design summary

Wall is 300 mm thick with 35 MPa concrete and 414 MPa steel

Vertical reinforcement is 20 mm dia at 125 mm spacing

Horizontal reinforcement is 12 mm dia at 150 mm spacing

Design status

PASS - wall is safe

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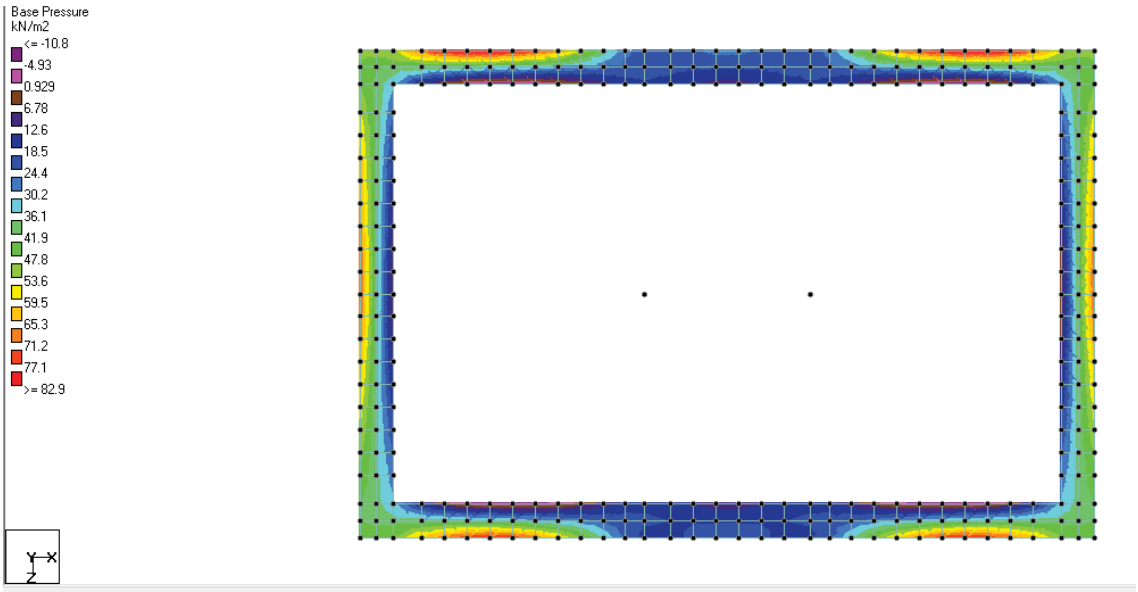
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12. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008 for Details)

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Explosive Magazine Area	BH-22	Explosive Magazine	Isolated Square	0.85	0.35 x 0.35	2000	225	225	25	27,000	20,900
				1.20	1.0 x 1.0	1400	440	440	25	52,800	32,200
				1.50		1800	600	600	25	72,000	38,900
				1.20	2.0 x 2.0	900	550	550	25	66,000	37,500
				1.50		1050	690	690	25	82,800	44,200
				1.20	3.0 x 3.0	700	650	650	25	78,000	42,800
				1.50		750	780	750	25	90,000	49,500
				1.20	4.0 x 4.0	575	800	575	25	69,000	48,100
				1.50		625	950	625	25	75,000	54,800
				1.20	5.0 x 5.0	500	950	500	25	60,000	53,400
				1.50		550	1110	550	25	66,000	60,200

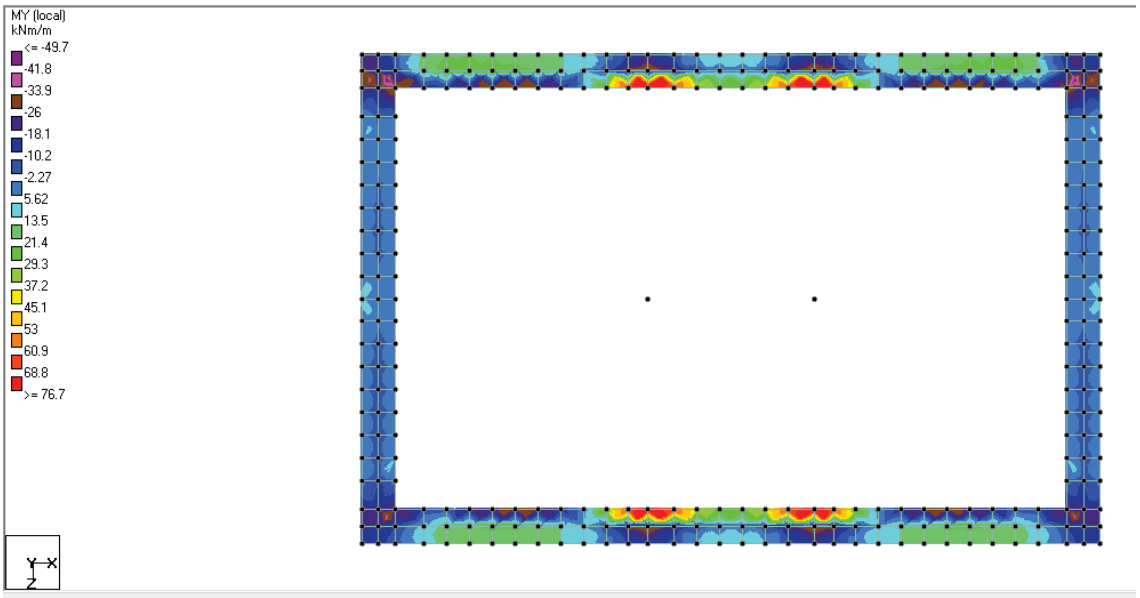


MAXIMUM PRESSURE (83 KPa < 350 KPa) UNDER THE STRIP FOOTING FROM STAAD RESULTS (DL+LL+BL)

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MAXIMUM MY IN THE STRIP FOOTING FROM STAAD RESULTS (1.2DL+1.6LL+BL)

```
      FY:    413.682 MPA FC:     35.000 MPA COVER (TOP):     19.050 MM
      COVER (BOTTOM):     19.050 MM TH:    500.000 MM
 4523 TOP : Longitudinal direction - Only minimum steel required.
 4523 BOTT: Longitudinal direction - Only minimum steel required.
 4523 TOP : Transverse direction   - Only minimum steel required.
 4523 BOTT: Transverse direction   - Only minimum steel required.
 4523 TOP :      0.900         21.64 /     5       0.900          78.91 /     5
      BOTT:      0.900          0.00 /     3       0.900           0.00 /     3
```

MAXIMUM AREA OF STEEL IN THE FOOTING

Provide Minimum Steel Reinf. of 904mm2/m

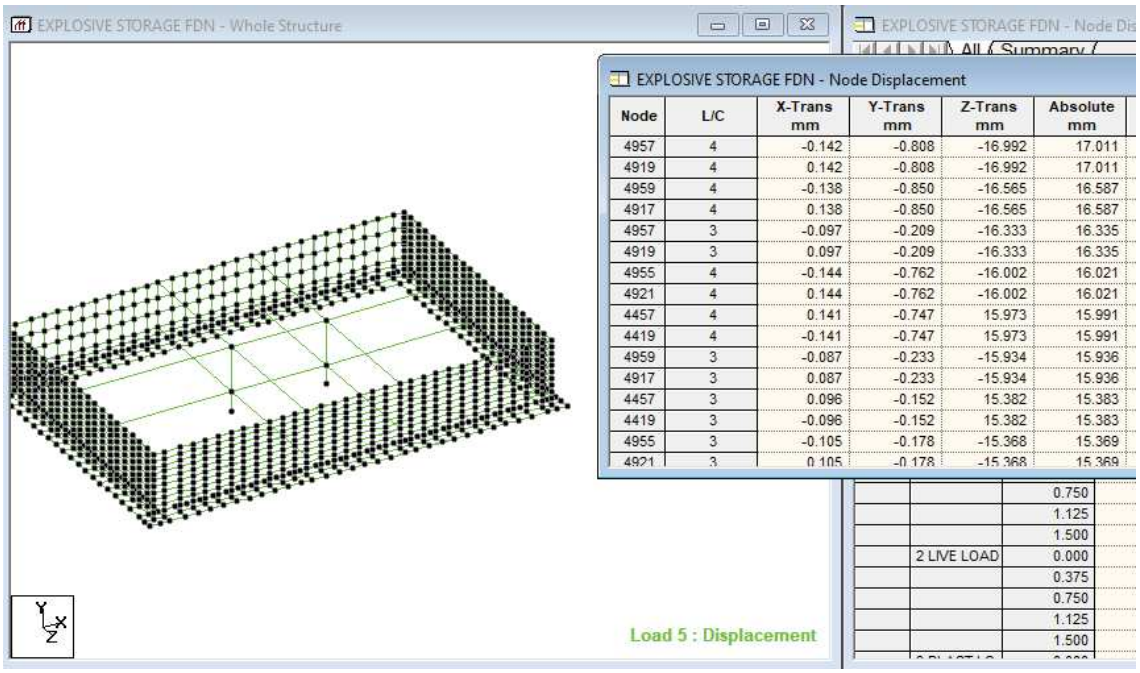
Note: The Foundation Sizes are subjected to change upon receiving of Final PEB Reactions and their Design Assumptions

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13. DRIFT CHECK



EXPLOSIVE STORAGE FDN - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm
4957	4	-0.142	-0.808	-16.992	17.011
4919	4	0.142	-0.808	-16.992	17.011
4959	4	-0.138	-0.850	-16.565	16.587
4917	4	0.138	-0.850	-16.565	16.587
4957	3	-0.097	-0.209	-16.333	16.335
4919	3	0.097	-0.209	-16.333	16.335
4955	4	-0.144	-0.762	-16.002	16.021
4921	4	0.144	-0.762	-16.002	16.021
4457	4	0.141	-0.747	15.973	15.991
4419	4	-0.141	-0.747	15.973	15.991
4959	3	-0.087	-0.233	-15.934	15.936
4917	3	0.087	-0.233	-15.934	15.936
4457	3	0.096	-0.152	15.382	15.383
4419	3	-0.096	-0.152	15.382	15.383
4955	3	-0.105	-0.178	-15.368	15.369
4921	3	0.105	-0.178	-15.368	15.369

		0.750
		1.125
		1.500
	2 LIVE LOAD	0.000
		0.375
		0.750
		1.125
		1.500

MAXIMUM DRIFT IN THE BUILDING IS 17 mm << 30mm (H/100)

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ATTACHMENT 1 COMMENT RESOLUTION SHEET

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-007785, 9 January 2021

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 9 January 2021	Comment by	EPC Response DATE: 15 January 2021	Status	Remarks
			General Comments (cc)				
MD-216-2500-EG-CV-CAL-2002_RA02	2500	1	Not sure if ASCE is complying to NFPA 495 code, Saudi Explosives Police Division, HCIS and US-ATF guidelines and directives. This section should highlight some applicable codes in Saudi Arabia.	Mr. Doron Golan (Bechtel)	ASCE code is reffered in HCIS -SAF-03 item 6.2 'blast resistance' and is also recommended by NFPA code to follow in blast design.. Applicable codes is metioned in item 4.1 in calculation document.	Closed	

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-004667, 3 September 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 3 September 2020	Comment by	EPC Response DATE: 14 December 2020	Status	Remarks
			General Comments (cc)				
MD-216-2500-EG-CV-CAL-2002_RA01	2500	1	This document forms part of the HCIS Stage 3 submission package and therefore shall be reviewed by the appointed SEC consultant KDOCS, Technical design shall follow HCIS SEC & SAF Directives, NFPA 495 code, and Explosive Police requirements related to explosives magazines. EPC shall refrain to submit individual reviews as it is not in compliance to HCIS workflow protocol, however this shall be submitted for review in a unified package ONLY. Similar observance to transmittals GAD- 2003, 2004, 2007, 2027, 2028	Mr. Doron Golan (BECHTEL)	Resubmitted for approval as advised by employer	Closed	